Covariant Approximate Quantum Codes for Protected Analog Computation

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Quantum error correction compatible with continuous symmetries is a fundamental problem in quantum information and a possible route to robust analog quantum simulation. Because the Eastin-Knill theorem forbids exact codes with continuous transversal symmetries, we construct explicit $SU(d)$-covariant approximate codes that exploit permutation symmetry to spread logical information uniformly across all physical subsystems. For one-, two-, and three-qudit erasures at known locations, we prove worst-case purified-distance scaling $O(1/N)$, which matches known approximate Eastin-Knill lower bounds up to constants, and we extend the reduced-state analysis to general flagged local noise. Exact permutation symmetry of this kind is available only up to three erased sites; for an arbitrary fixed number k of erased sites, we instead show that Haar-random and efficiently sampled encodings achieve worst-case distance $O(\sqrt{k}/N)$ with probability exponentially close to one. For single-qudit erasure, we construct an explicit near-optimal decoder from the Petz recovery map. We then use these codes as building blocks for encoded analog dynamics. Symmetry-preserving Hamiltonians generate block-structured dynamical Lie algebras implementable transversally, while controlled symmetry-breaking terms serve as non-transversal resources for universal dynamics. These results provide explicit non-Abelian covariant codes and a framework for robust analog quantum simulation.

I. INTRODUCTION

Digital quantum computing has made substantial progress. Programmable processors have reached regimes that are hard to simulate classically, with growing evidence for useful pre-fault-tolerant computations in specific settings [1, 2]. These experiments do not yet use full quantum error correction, but digital quantum computing has a clear fault-tolerance paradigm: encode logical qubits into many physical qubits, extract error information and correct errors repeatedly, and implement protected logical gates. Although the overheads remain large [3, 4], the conceptual route to long fault-tolerant digital algorithms is in place. Analog platforms offer a complementary route to quantum simulation and computation [5–8], with recent experiments preparing strongly correlated states, including Fermi-Hubbard systems, in regimes challenging for classical methods [9–12]. More generally, analog devices implement continuously generated Hamiltonian dynamics, and universal analog computation can be formulated by asking whether the available Hamiltonians generate a sufficiently large dynamical Lie algebra (DLA) [13–22]. The promise of analog computation is therefore clear, but the corresponding framework for protecting native continuous-time dynamics is much less developed than in the digital circuit model.

Noise control is central in both settings. Digital fault tolerance is supported by a mature theory and by rapid experimental progress in logical memories and small logical processors, including break-even demonstrations where increasing the number of physical qubits reduces logical error rates [23–29]. For analog simulation, one can in principle digitize the target evolution and run it in a protected circuit, but this replaces continuous Hamiltonian evolution by long gate sequences and adds Trotterization or phase-estimation costs to the fault-tolerance overhead [30, 31]. This motivates noise-control strategies that are native to analog dynamics. Recent work has identified noise-stable regimes with quantum advantage [32], provided accuracy and hardness guarantees for noisy open-system simulations [33], and proposed direct error-reduction methods based on Hamiltonian reshaping or rescaling [34], penalty-Hamiltonian encodings [35], and continuous syndrome monitoring with feedback [36]. These ideas show that analog noise control need not simply reproduce digital fault tolerance. What is still missing is a general encoding principle that suppresses local noise while preserving the continuous encoded dynamics that make analog platforms natural.

A fundamental obstruction is the Eastin–Knill theorem [37]. In an encoded analog device, the natural fault-tolerant way to implement a continuous logical operation is to realize its generator transversally, as a sum of local physical generators across the code block. Equivalently, the desired logical dynamics can be viewed as a continuous group of logical transformations, and a transversal implementation realizes the same group on the physical subsystems by a product action. In this sense, the transversal implementation defines a continuous symmetry of the encoded system: applying the group action before encoding should be equivalent to encoding first and then applying the corresponding physical action. This is the *covariance* condition. However, exact finite-dimensional quantum error-correcting codes that correct local errors cannot support nontrivial continuous transversal symmetries, as stated in the Eastin–Knill theorem [37]. Thus continuous analog dynamics and exact transversal quantum error correction are in tension. One can relax this obstruction by using infinite-dimensional

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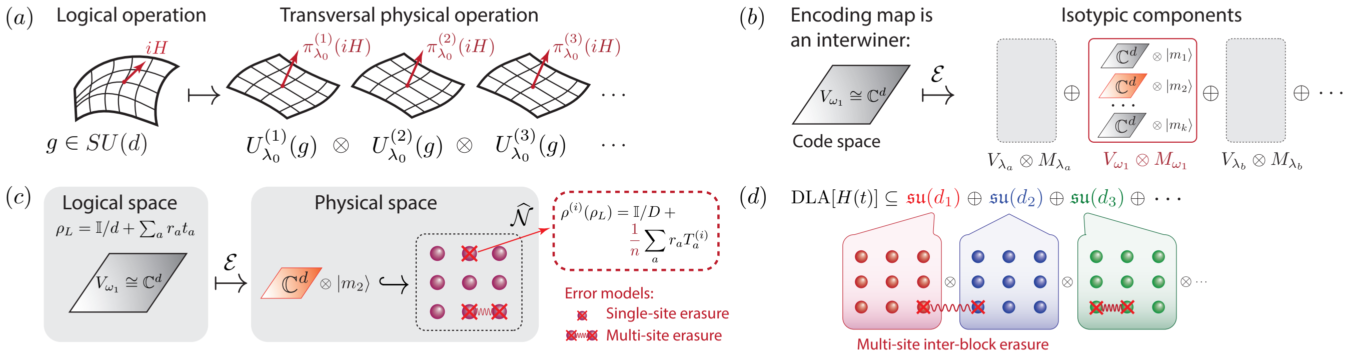

Figure 1. Overview of the covariant encoding construction and its applications. (a) Covariance relates the logical $SU(d)$ action to a physical transversal action. The logical generator iH is represented physically as a sum of local generators acting on the n physical subsystems, each carrying the same single-site representation λ_0. (b) The encoding isometry embeds the logical space $V_{\omega_1} \cong \mathbb{C}^d$ into the physical Hilbert space. After decomposing the physical space into isotypic components $V_\lambda \otimes M_\lambda$, covariance forces the code to lie inside the component $V_{\omega_1} \otimes M_{\omega_1}$. Thus, choosing the covariant code reduces to choosing a multiplicity vector $|m\rangle \in M_{\omega_1}$, which selects one embedded copy of the logical representation. (c) Error protection is analyzed through the complementary channel, which describes the information leaked to the environment. For a single-site erasure, the environment sees the reduced state $\rho^{(i)}(\rho_L)$ of the erased subsystem. In our codes, the logical-state-dependent part of this reduced state is suppressed as the number of physical subsystems grows, so the environment output becomes close to a constant channel. (d) The same codes can be used block-wise for symmetry-constrained analog simulation. The invariant part of the dynamical Lie algebra acts transversally within each encoded block, while multi-site erasures, including erasures involving different blocks, are treated using the same reduced-state and complementary-channel method.

covariant codes [38], by keeping exact finite-dimensional codes while restricting the correctable errors [39], or by allowing a controlled recovery error, leading to approximate quantum error correction (AQEC) [40, 41]. We follow the third route: finite-dimensional covariant codes with approximate correctability.

Covariant codes can support continuous groups of transversal logical operations at the cost of controlled approximation error [42]. Quantitative trade-offs between covariance and correctability are now known, including information-theoretic lower bounds for covariant codes under erasure noise [42–46]. Near-optimal scaling has been obtained from symmetric random constructions, and explicit examples have been developed for $U(1)$ covariance, W-state-type codes, and related questions in encoding complexity [42, 47–49]. These works establish the basic principles and limits of covariant AQEC, but many constructions are randomized, use reference frames or ancilla systems [45, 46], or address specific code families. Covariant approximate quantum error-correcting codes (AQECCs) are also relevant beyond computation. They model the protection of quantum information carrying a physical symmetry charge; they arise naturally in holographic quantum error correction and in discussions of approximate global symmetries in quantum gravity [50–52]; and they are closely connected to quantum reference frames and metrology, where the same constraints can be expressed through reference-frame accuracy or quantum Fisher information [38, 53, 54]. Explicit non-Abelian covariant AQECCs therefore provide concrete models for studying how symmetry, locality, and approximate correctability can coexist.

Covariant AQEC is therefore a natural candidate for protecting analog dynamics. Rather than digitizing the evolution into a long protected circuit, one chooses an encoded subspace that carries the required continuous evolution and suppresses the information that local noise leaks to the environment. Approximate correctability is what makes this possible in finite dimensions: it evades the Eastin–Knill obstruction while keeping transversal implementations of the continuous symmetry generators. This matters beyond analog computation, because the same trade-off between symmetry and correctability constrains any fault-tolerant scheme that has to respect a continuous symmetry. Explicit and flexible constructions, however, remain scarce. Many existing approaches assume Abelian or otherwise restricted symmetries, rely on randomization, or treat only single-site noise, whereas physical settings usually involve non-Abelian symmetries and errors acting on several subsystems or with a more general local structure. This leads to the question we address here: can one construct finite-dimensional covariant AQECCs for non-Abelian symmetries such as $SU(d)$, with controllable performance under general multi-qudit noise models?

We answer this question positively. As summarized in Section II, we construct explicit finite-dimensional $SU(d)$-covariant AQECCs in which the logical system transforms as the fundamental representation and the physical system consists of many local $SU(d)$ degrees of freedom. The main idea is to use the freedom available inside the physical representation space to impose permutation symmetries on the code. These symmetries spread the logical information nearly uniformly across the physical subsystems, so that the reduced states seen by local noise depend only weakly on the logical input. We first illustrate this mechanism for $SU(2)$ in Section IV, and then develop the general $SU(d)$ construction in Section V. This gives $O(1/n)$ worst-case purified-distance scaling for one-, two-, and three-qudit flagged erasures. For single-qudit erasure this scaling is near-optimal, matching known covariant-code lower bounds up to constants [42, 43]. Making all k-site reduced states exactly equal requires a permutation symmetry that can carry any k sites onto any other k sites, and no such symmetry is available once k exceeds three, so this route stops at three erased sites. In Section V C we reach an arbitrary fixed number k of erased sites by replacing exact symmetry with typicality. Covariance alone leaves a large family of admissible encodings, and we show that a Haar-random member of this family is, with probability exponentially close to one in n, simultaneously good for all $\binom{n}{k}$ erased sets, with worst-case distance $O(\sqrt{k(d^2-1)}/n)$. Read as a statement about covariant codes as a whole, this says that the encodings leaking an atypically large amount of logical information to some erased subsystem form an exponentially small fraction of the family. Typicality by itself does not produce a code one can write down, because specifying a Haar-random encoding takes exponentially many parameters, and sampling one costs as many random bits. We therefore narrow the search to a finite subfamily of encodings, each generated from a short seed. Sampling from this subfamily uses $O(n \log d)$ random bits, every parameter of the sampled encoding is computable in $\mathrm{poly}(n, \log d)$ time, and the resulting code meets the same $O(\sqrt{k(d^2-1)}/n)$ bound with probability exponentially close to one. The same reduced-state method also gives general $O_{d,k}(n^{-1/2})$ bounds for arbitrary flagged local noise. For single-qudit erasure, we additionally give an explicit near-optimal decoder based on the Petz recovery map [55–57]. In Section VI we use these codes as building blocks for analog quantum simulation. A simulator does not need every unitary on its Hilbert space, since the evolutions it can reach form the dynamical Lie algebra generated by its drift and control Hamiltonians, which is typically much smaller than the algebra of all Hamiltonians on that system. We therefore build an encoding covariant only with respect to this smaller algebra, so that every Hamiltonian the simulator can reach is implemented transversally on the physical qudits. When the available Hamiltonians share a site-permutation symmetry, the algebra splits into blocks, and we encode each block separately, reusing the codes above together with known $U(1)$-covariant constructions [47]. The same $1/n$ protection persists for the two-qudit erasure models analyzed here, including erasures that involve qudits from different blocks. Finally, in Section VII, we explain how controlled symmetry-breaking Hamiltonians can be used as non-transversal resource operations to enlarge the protected symmetry-preserving dynamics to universal analog control. Proofs, the nomenclature, and the supporting calculations are collected in the Supplemental Material [58]. Results stated there carry the prefix S, so that Lemma S3.1 and Eq. (S17) always refer to the Supplemental Material rather than to this article.

II. SUMMARY OF MAIN RESULTS

The following sections derive the main results using a representation-theoretical framework. This section summarizes them with minimal use of that formalism, for readers interested in the formulas and bounds rather than the derivations. The subsections are presented in the same order as the corresponding sections of the paper.

A. Approximate error correction framework

A code is an isometry $\mathcal{V}$ embedding of a small logical Hilbert space $\mathcal{H}_L$ into a large physical Hilbert space $\mathcal{H}_P = A_1 \otimes \cdots \otimes A_n$ of n qudits, $\mathcal{E}(\rho_L) := \mathcal{V}\rho_L\mathcal{V}^\dagger$. We say the code ε-approximately protects against a noise channel $\mathcal{N}$ if some recovery channel $\mathcal{R}$ can undo the noise approximately,

$$d(\mathcal{R} \circ \mathcal{N} \circ \mathcal{E}, \mathrm{id}_L) \leq \varepsilon,$$

where d is a fidelity-based distance between channels, Eq. (1). For a family of codes indexed by n (usually being the number of physical qudits), the question is how fast ε shrinks as n grows. The main noise model is erasure at known locations, i. e. with probability p_S, the qudits in a set S are discarded and replaced by a blank state, with a flag recording which qudits were lost, Eq. (2); we also treat the more general case of an arbitrary, possibly unknown error acting on a known set of qudits, Eq. (3).

A convenient way to think about correctability, due to Bény and Oreshkov [41], is to ask what an environment having access only to the discarded qudits could learn about the logical state. If that environment's state is essentially the same no matter what was encoded, the logical information was never really leaked, and a good recovery map is guaranteed to exist, Eq. (5). The relevant object is therefore the reduced state of the encoded qudits seen by this observer,

$$\rho^{(S)}(\rho_L) := \mathrm{Tr}_{\hat{S}}\big(\mathcal{V}\rho_L\mathcal{V}^\dagger\big)\,,$$

and most of the technical work in the paper is a calculation of $\rho^{(S)}(\rho_L)$ for small sets S of one, two, or three qudits, followed by an estimate of how strongly it depends on ρ_L.

The approximate codes we build are also required to satisfy a precise covariance condition with respect to a continuous symmetry $SU(d)$ (or a product of such groups). Let $U_L(g)$ denote the action of $g \in SU(d)$ on the logical space, and let $U_P(g)$ denote the transversal physical action obtained by applying one fixed local rotation to every qudit at once, Eq. (15). Covariance means that the encoding isometry $\mathcal{V}$ satisfies

$$\mathcal{V}U_L(g) = U_P(g)\mathcal{V}, \qquad \forall g \in SU(d),$$

Eq. (10). This means that rotating the logical state and then encoding it gives exactly the same physical state as encoding it first and then rotating every qudit transversally. Thus, an approximate quantum error-correcting code satisfying the covariance condition provides a code space on which the logical $SU(d)$ symmetry is implemented by continuous transversal physical operations.

Equation (10) is a strong constraint, because it forces $\mathcal{V}$ to map the logical space onto one specific copy of itself sitting inside the physical Hilbert space, among the several equivalent copies that occur there once n is large enough. What this constraint does not fix is *which* copy to choose when more than one is available, and this is exactly the freedom we exploit. We pick the copy so that it is also left invariant by permuting the physical qudits, which forces the logical information to be shared equally among them; an asymmetric choice, by contrast, is exactly what would let a handful of erased qudits give away a disproportionate share of the logical information.

B. A first example: one logical qubit shared among many spins

The simplest version of the construction encodes a logical spin-$\frac{1}{2}$ qubit into n physical spin-j systems (n odd, j half-integer), with logical $SU(2)$ rotations implemented as the total spin $J_a = \sum_k J_a^{(k)}, a = x, y, z$. Requiring the code to be invariant under cyclic permutations of the n spins forces the logical information to be shared exactly equally among them. The reduced state of any single spin is then

$$\rho^{(i)}(\rho_L) = \frac{I_{2j+1}}{2j+1} + \frac{3}{2nj(j+1)(2j+1)} \sum_a r_a J_a^{(i)},$$

for $\rho_L = \frac{1}{2}(I + \mathbf{r} \cdot \boldsymbol{\sigma})$. The first term, maximally mixed, carries no information about ρ_L; the second term, which does, is suppressed by $1/n$. Translating this into a recovery bound for flagged single-spin erasure gives

$$d(\mathcal{RNE}, \mathrm{id}) \leq \frac{3}{4\sqrt{2j(j+1)}} \frac{1}{n} + O_j(n^{-2}).$$

Known lower bounds for any $SU(2)$-covariant code under erasure scale the same way in n [42], so this code is near-optimal in its n-scaling.

C. The general $SU(d)$ code, a decoder, and general noise

The same mechanism works for a d-dimensional logical system encoded into n copies of a fixed local representation of $SU(d)$ (each physical qudit carries an antisymmetric tensor power $V_{\omega_r} = \bigwedge^r \mathbb{C}^d$ of the fundamental representation, with generators acting as $T_a^{(i)}$; Eq. (39)). The cyclically symmetric code again gives a one-qudit reduced state of "maximally mixed plus a $1/n$-suppressed, logical-dependent piece,"

$$\rho^{(i)}(\rho_L) = \frac{I_{V_{\omega_r}}}{\dim V_{\omega_r}} + \frac{1}{2\kappa_{\omega_r} n} \sum_a r_a T_a^{(i)}, \quad \kappa_{\omega_r} = \frac{1}{2}\binom{d-2}{r-1},$$

and a single-qudit erasure bound

$$d(\mathcal{RNE}, \mathrm{id}) \leq \frac{d-1}{2\sqrt{2}} \sqrt{\frac{d+1}{r(d-r)}} \frac{1}{n} + O_d(n^{-2}),$$

which again matches the known $SU(d)$ lower bound [42] in its $1/n$ scaling, up to a constant depending on d and r.

We then provide an explicit recovery map, the Petz (transpose) map built directly from the encoding isometry $\mathcal{V}$, and show that it achieves the same $1/n$ scaling, though we do not pin down its optimal constant.

The same reduced state also controls protection against non-erasure noise. If a single qudit undergoes an arbitrary unknown local noise channel, then the environment does not see $\rho^{(i)}(\rho_L)$ directly, but only its image under the corresponding local complementary channel. Thus, all information about the logical state still passes through the one-site reduced state $\rho^{(i)}(\rho_L)$ above. Since the logical-state-dependent part of this reduced state is suppressed with n, the code remains approximately correctable, with the weaker scaling $O_d(1/\sqrt{n})$. The loss from $1/n$ to $1/\sqrt{n}$ comes from using a fully general, but less tight, proof technique. The same argument covers noise on several qudits at once, so we treat both cases together in Section V D.

D. Protecting against multi-qudit noise

If two or three qudits can be erased simultaneously, at unknown but flagged locations, cyclic symmetry is no longer enough: every pair, respectively every triple, of qudits must look alike to the environment. This requires the code to be invariant under a subgroup of the permutation group that can move any pair (or triple) of sites to any other pair (or triple) – a 2- or 3-transitive subgroup of all permutations of the n sites. Such subgroups are rare; we use the affine group $\mathrm{AGL}(1, n)$, Eq. (46), for two-qudit erasures and the projective linear group $\mathrm{PGL}(2, n-1)$, Eq. (47), for three-qudit erasures, both of which admit the needed invariant codes once n (respectively $n-1$) is a suitable prime power. (The full permutation group S_n would also work in principle, but it is too restrictive, as we demonstrate in the corresponding section.)

With this extra symmetry, the three-qudit reduced state splits into a logical-independent piece and a logical-dependent piece of size $\Theta(1/n)$, $\rho^{(ijk)}(\rho_L) = \tau^{(ijk)} + \Delta^{(ijk)}(\rho_L)$, Eq. (48), and the resulting bound for flagged three-qudit erasure is

$$d(\mathcal{RNE}, \mathrm{id}) \leq \frac{\sqrt{3(d^2-1)}}{2\sqrt{2}} \frac{1}{n} + O_d(n^{-2}).$$

As in the single-qudit case, the same reduced state controls protection against arbitrary (non-erasure) noise on up to three known qudits, at the weaker rate $O_d(1/\sqrt{n})$.

E. Erasure on an arbitrary fixed number of qudits

The construction above stops at three erased sites because the k-transitive subgroups of S_n needed to force all k-site reduced states to coincide exactly are not available for $k > 3$. Apart from the alternating group, which is so large that no covariant code is invariant under it, the classification of finite multiply transitive permutation groups leaves only the Mathieu groups, and these occur at four values of n and so support no construction with $n \to \infty$. To go beyond three sites, we drop exact equality of the k-site reduced states and ask only that they be close to one another, for a well-chosen code.

Closeness has to be measured against something, and the natural reference is an average over a whole family of codes. We start from a simple seed encoding that places the logical qudit at one site and fills the remaining sites with independent $SU(d)$-singlets, Eq. (S44). The covariance condition leaves a whole family of admissible encodings, Eq. (S47), and we average over all of them.

This average, taken with respect to the Haar measure on that family, defines the mean k-site channel. This channel belongs to no single encoding. It is the uniform mixture of all covariant encodings, and Proposition S3.37 identifies that mixture with the seed encoding twirled over the full permutation group S_n. This is what averaging buys. No single member of the family is invariant under all of S_n, for the same reason that ruled out the alternating group above, but the mixture is. Its k-site reduced state is therefore the same for every set of k sites, and it stays close to maximally mixed up to a logical-dependent correction of order $1/n$, just as in the one-, two-, and three-site cases above. The mean channel is thus the common point around which well-chosen codes should cluster.

A concentration argument (Theorem 8) then shows that a single Haar-random draw from this family is simultaneously close to the mean channel for every one of the $\binom{n}{k}$ possible erased sets, with probability approaching one exponentially fast in n. Drawing exactly from the family would require exponentially many random bits, since specifying one of its members takes exponentially many parameters; we instead give an explicit randomness-efficient sampling procedure (Theorem 10) that reproduces the same guarantee from a seed of only $O(n \log d)$ random bits.

A Haar-random code already achieves the resulting erasure bound (Theorem 9), and combining it with the efficient sampling gives a family of $SU(d)$-covariant codes, correctable with high probability over the random seed against flagged erasure on any fixed number k of sites, uniformly over the distribution of erased sets (Corollary 1):

$$d(\mathcal{R}_p \mathcal{N}\mathcal{E}, \mathrm{id}) \leq \frac{\sqrt{k(d^2-1)}}{2\sqrt{2}} \frac{1}{n} + O_{d,k}(n^{-2}).$$

The $1/n$ protection against flagged erasure is therefore not limited to at most three sites, and it extends, with the same scaling in n and an explicit $\sqrt{k}$ dependence on the number of erased sites, to erasure on any fixed number of physical sites. We make no optimality claim for $k > 1$, since the covariant lower bounds we use [42, 43] are stated for single-qudit erasure, so neither the $1/n$ constant nor the $\sqrt{k}$ dependence is matched by a converse for k-site erasure.

This also changes how the construction should be read. Covariance by itself leaves the whole family of admissible encodings, and what the concentration argument says is that the members for which some erased subsystem reveals an atypically large amount of logical information form an exponentially small fraction of it. Poorly performing encodings do exist, the seed encoding itself among them, since erasing the one site that carries the logical qudit reveals the logical state completely. But they are non-generic, so the protection does not come from fine-tuning one special code, it becomes clear that approximate decoupling from any fixed-size erased subsystem is a typical property of covariant codes. The efficient sampler then shows that such an encoding can also be built. Full Haar randomness is not what the error-correcting property needs: the concentration argument uses only second and fourth moments, so four-wise independent phases generated from a short seed do the same work.

F. Using these codes to protect analog quantum dynamics

We also use this machinery to protect continuous-time (analog) quantum simulation under a symmetry. If the Hamiltonians driving an analog simulator are invariant under some finite group G of qubit permutations, every reachable evolution is confined to a block-diagonal invariant algebra

$$\mathfrak{su}(\mathcal{H})^G \cong \mathfrak{su}(d_1) \oplus \cdots \oplus \mathfrak{su}(d_k) \oplus \mathfrak{u}(1)^{\oplus(r-1)},$$

Eq. (54), where the block sizes d_i are fixed by G and are typically far smaller than the full Hilbert-space dimension. We then build a *block encoding*, Eq. (56): each non-Abelian block $\mathfrak{su}(d_i)$ is encoded separately, using one of the $SU(d_i)$-covariant codes above, so that every symmetry-respecting Hamiltonian acts transversally, block by block, on the encoded system.

This block code still tolerates noise that mixes different blocks. For a two-block code (block sizes n_1, n_2) under flagged two-qudit erasure – whether both erased qudits lie in the same block or one in each – the recovery error again scales as

$$d(\mathcal{RNE}, \mathrm{id}) \leq \frac{\sqrt{C_1(d_1^2-1)+C_2(d_2^2-1)}}{n} + O_{d_1,d_2}(n^{-2})$$

for $n_1 \approx n_2 \approx n$, Eq. (61), with constants C_1, C_2 independent of the block dimensions. Stacking covariant codes block by block therefore preserves the same $1/n$ protection, even for erasures affecting two blocks.

G. Towards universal analog computation

Symmetric Hamiltonians alone can only ever generate the smaller, block-diagonal invariant algebra above, which in general doesn't coincide with the full algebra needed for universal control. We give a sufficient condition, Theorem 14, under which adding a modest set of *symmetry-breaking* Hamiltonians restores universality: if the resulting "coupling graph" linking the different symmetry blocks is connected, and each link is "rich enough" in a precise sense (the Full First-Factor Span condition of Appendix S5), then the symmetric Hamiltonians together with the symmetry-breaking ones generate the full algebra $\mathfrak{su}(\mathcal{H})$ on the whole Hilbert space. For instance, three qubits with S_3 permutation symmetry become fully controllable once a single extra term, $Z_1 + X_2$, is added to the symmetric Hamiltonians.

This suggests a possible route toward robust universal analog computation, rather than a complete architecture by itself. The block encoding protects the encoded code space, within which symmetry-preserving Hamiltonians can be implemented transversally on the corresponding covariant code blocks. Universality then requires additional symmetry-breaking Hamiltonians, which are not transversal under the block encoding and therefore act as resource operations. The usefulness of this approach depends on choosing a symmetry that gives a favorable tradeoff: the invariant block structure should reduce the required physical resources, such as the number of physical qudits or their local dimensions, while the required symmetry-breaking resource Hamiltonians should remain sufficiently simple to implement.

III. PRELIMINARIES

A. Approximate Quantum Error Correction

We first recall the approximate error-correction framework used throughout the paper. An exact quantum error-correcting code for a given noise model is an encoding for which there exists a recovery channel that perfectly restores the logical information after the noise has acted. This requirement is too restrictive for the setting considered here. In particular, the Eastin–Knill theorem and its variants imply that finite-dimensional codes correcting local errors exactly cannot support nontrivial continuous logical symmetries transversally [37].

Approximate quantum error correction relaxes this condition. For a fixed noise channel, an encoding is ε-approximately correctable if there exists a recovery channel that reverses the noisy encoded evolution up to error at most ε, calculated by a chosen distance between quantum channels [41]. For an asymptotic family of codes, the central requirement is that this error decreases as the number n of physical subsystems grows.

The noise model has to be specified with some care. If the noise is artificially weak, approximate correctability can arise for trivial reasons. For example, if exactly one physical qubit is erased uniformly at random among n qubits, then the trivial encoding

$$|\psi\rangle \longmapsto |\psi\rangle\, |0\rangle^{\otimes(n-1)}$$

has vanishing error as $n \to \infty$, simply because the logical qubit is erased only with probability $1/n$ [49]. This does not constitute a robust error-correction mechanism. Thus, throughout the paper, we consider asymptotic code families for which the error parameter ε decreases with the system size, and we work with the most general noise models for which the performance can be analyzed explicitly.

For a fixed code and noise model, one could in principle optimize over all recovery channels. This optimization is generally difficult [59]. We instead use the complementary-channel characterization of approximate error correction. The complementary channel describes the information leaked to the environment. Approximate correctability is equivalent to the statement that this environment output is close to independent of the logical input, or equivalently that the complementary channel is close to a constant channel, i.e., a quantum channel that maps all states to some fixed state [41, 42].

a. Correctability and distance. We now define the channel distance used in the paper. For two quantum channels $\mathcal{N}$ and $\mathcal{M}$, their worst-case entanglement fidelity is

$$F(\mathcal{N},\mathcal{M}) := \min_{\rho} f\Big((\mathcal{N}\otimes\mathrm{id})(|\psi_\rho\rangle\langle\psi_\rho|), (\mathcal{M}\otimes\mathrm{id})(|\psi_\rho\rangle\langle\psi_\rho|)\Big),$$

where $|\psi_\rho\rangle$ is a purification of ρ, and

$$f(\rho,\sigma) := \mathrm{Tr}\sqrt{\sqrt{\rho}\sigma\sqrt{\rho}}$$

is the root fidelity between states. We use the corresponding fidelity-induced channel distance

$$d(\mathcal{N},\mathcal{M}) := \sqrt{1 - F(\mathcal{N},\mathcal{M})}. \tag{1}$$

Let

$$\mathcal{E} : \mathcal{D}(\mathcal{H}_L) \to \mathcal{D}(\mathcal{H}_P)$$

be an encoding channel, where $\mathcal{H}_L$ and $\mathcal{H}_P$ are the logical and physical Hilbert spaces. For a fixed noise channel $\mathcal{N}$, we say that $\mathcal{E}$ is ε-correctable under $\mathcal{N}$ [41] if there exists a recovery channel $\mathcal{R}$ such that

$$d(\mathcal{R}\circ\mathcal{N}\circ\mathcal{E}, \mathrm{id}_L) \leq \varepsilon.$$

b. Flagged local noise. Throughout the paper, the physical Hilbert space of n qudits is decomposed as

$$\mathcal{H}_P = A_1 \otimes \cdots \otimes A_n,$$

where A_i is the Hilbert space of the ith qudit. For a subset $S \subseteq 1, \ldots, n$, we write

$$A_S := \bigotimes_{i \in S} A_i,$$

and $\hat{S}$ denotes the complementary subset. The hat denotes complementation throughout: $\widehat{ijk}$ for the complement of $\{i, j, k\}$, $\widehat{j}$ for the retained tensor factors other than j, and $\widehat{\mathcal{N}}$ for the complementary channel of $\mathcal{N}$. To simplify notation, when $S = i, j, k$, we write A_{ijk}, Tr_{ijk}, p_{ijk}, and $|ijk\rangle$ instead of A_S, Tr_{A_S}, p_S, and $|S\rangle$. Similarly, for a single site $S = i$, we often write i instead of A_i.

The main noise model considered in the paper is subsystem erasure at known locations, or equivalently flagged subsystem erasure. In its general form, it is

$$\mathcal{N}(\sigma) = \sum_S p_S \left|S\right\rangle \left\langle S\right|_F \otimes \left|e\right\rangle \left\langle e\right|_{A_S} \otimes \mathrm{Tr}_S(\sigma), \quad (2)$$

where F is a classical flag register recording which subsystem A_S was erased [42]. We also consider an arbitrary flagged multi-qudit noise of the form

$$\mathcal{N}(\sigma) = \sum_{S \in \mathcal{S}} p_S \left|S\right\rangle \left\langle S\right|_F \otimes \left(\mathcal{N}_S \otimes \mathrm{id}_{\hat{S}}\right)(\sigma), \quad (3)$$

where $\mathcal{N}_S$ is an arbitrary noise channel acting on the subsystem $A_S = \bigotimes_{i \in S} A_i$, extending beyond erasure errors.

c. Complementary-channel criterion. Let

$$\Phi : \mathrm{End}(\mathcal{H}_A) \to \mathrm{End}(\mathcal{H}_B)$$

be a quantum channel with Stinespring dilation

$$\Phi(\sigma) = \mathrm{Tr}_E(W \sigma W^\dagger), \qquad W : \mathcal{H}_A \to \mathcal{H}_B \otimes \mathcal{H}_E.$$

The complementary channel is obtained by tracing out the channel output:

$$\widehat{\Phi}(\sigma) := \mathrm{Tr}_B(W \sigma W^\dagger).$$

Thus Φ describes the system output, while $\widehat{\Phi}$ describes the corresponding environment output.

For a state ρ_0 on the environment output space, the corresponding constant channel is

$$\Lambda_0(\rho) := \mathrm{Tr}(\rho)\rho_0. \quad (4)$$

The following theorem relates approximate correctability to decoupling from the environment.

Theorem 1 (Bény and Oreshkov [41]). *Let $\mathcal{E}$ be an encoding channel, let $\mathcal{N}$ be a noise channel, and let $\widehat{\mathcal{N} \circ \mathcal{E}}$ be a complementary channel to $\mathcal{N} \circ \mathcal{E}$. Then*

$$\min_{\mathcal{R}} d\left(\mathcal{R} \circ \mathcal{N} \circ \mathcal{E}, \mathrm{id}_L\right) = \min_{\Lambda_0\,\mathrm{constant}} d\left(\widehat{\mathcal{N} \circ \mathcal{E}}, \Lambda_0\right). \quad (5)$$

The theorem states that an encoding is approximately correctable precisely when the environment output can be made close to a fixed state that is independent of the logical input. Therefore, throughout the paper rather than constructing a recovery map first, we control what the environment can learn.

d. Reduced states seen by the environment. For the erasure channel, defined in Eq. (2), suppose that the encoding is isometric:

$$\mathcal{E}(\rho_L) = \mathcal{V} \rho_L \mathcal{V}^\dagger. \quad (6)$$

Hence, a complementary channel to the erasure channel, derived in [42], is

$$\widehat{\mathcal{N} \circ \mathcal{E}}(\rho_L) = \sum_S p_S \left|S\right\rangle \left\langle S\right|_{F_E} \otimes \mathrm{Tr}_{\hat{S}}\left(\mathcal{V} \rho_L \mathcal{V}^\dagger\right). \quad (7)$$

Thus, the environment receives the erased subsystem, together with a flag recording which subsystem was erased.

For the arbitrary flagged multi-qudit noise model (3), let $\widehat{\mathcal{N}}_S$ be a complementary channel to $\mathcal{N}_S$. Then, as shown in Lemma S3.48,

$$\widehat{\mathcal{N} \circ \mathcal{E}}(\rho_L) = \sum_{S \in \mathcal{S}} p_S \left|S\right\rangle \left\langle S\right|_{F_E} \otimes \widehat{\mathcal{N}}_S \left(\mathrm{Tr}_{\hat{S}}\left(\mathcal{V} \rho_L \mathcal{V}^\dagger\right)\right). \quad (8)$$

Eqs. (7) and (8) show that local reduced states of the encoded state are the central objects in the complementary-channel analysis. We therefore use the notation

$$\rho^{(S)}(\rho_L) := \mathrm{Tr}_{\hat{S}}\left(\mathcal{V} \rho_L \mathcal{V}^\dagger\right), \quad (9)$$

where $S \subseteq 1, \ldots, n$. In particular,

$$\rho^{(i)}(\rho_L), \qquad \rho^{(ij)}(\rho_L), \qquad \rho^{(ijk)}(\rho_L)$$

denote the one-, two-, and three-qudit reduced states, respectively. The main calculations below are calculations of these reduced states and of how much they depend on the logical input.

B. Covariant encoding

We now introduce the covariant encoding framework used throughout the paper. Approximate correctability allows us to keep continuous logical symmetries implemented locally, while relaxing exact correction to correction with a controlled error. The compatibility condition between an encoding and a symmetry is *covariance* [38]. In representation-theoretic language, covariance says that the encoding map is an *intertwiner* between the logical and physical representations.

For the codes constructed below, the logical system carries the fundamental representation of $SU(d)$. Constructing a covariant code therefore amounts to embedding a copy of this representation into the physical Hilbert space. Representation theory identifies where such copies can occur, and the remaining freedom is a choice of vector in the corresponding multiplicity space [60]. We use this freedom to impose additional permutation symmetries on the code space. Specifically, we choose the multiplicity vector to be invariant under a suitable subgroup of the permutation group on n elements $G_P \subseteq S_n$ acting by permutations of the physical subsystems. Since this permutation action commutes with the collective $SU(d)$ action, it acts only on multiplicity spaces. Thus, a G_P-invariant multiplicity vector gives a G_P-invariant code space. The choice of G_P depends on the noise model and will be specified in the later sections.

a. Covariance. Let G be a compact connected Lie group, and let $\mathfrak{g}$ be its Lie algebra. Let

$$\mathcal{V} : \mathcal{H}_L \to \mathcal{H}_P$$

be the encoding isometry associated with the encoding channel in Eq. (6). Let U_L and U_P be unitary representations of G on $\mathcal{H}_L$ and $\mathcal{H}_P$, respectively. We say that $\mathcal{V}$ is *G-covariant with respect to U_L and U_P* if

$$\mathcal{V} U_L(g) = U_P(g) \mathcal{V}, \qquad \forall g \in G. \tag{10}$$

Equivalently, let π_L and π_P be the corresponding Lie-algebra representations,

$$\pi_L(X) := \frac{d}{dt}\bigg|_{t=0} U_L(e^{tX}), \qquad \pi_P(X) := \frac{d}{dt}\bigg|_{t=0} U_P(e^{tX}).$$

Then $\mathcal{V}$ is *$\mathfrak{g}$-covariant with respect to π_L and π_P* if

$$\mathcal{V} \pi_L(X) = \pi_P(X) \mathcal{V}, \qquad \forall X \in \mathfrak{g}. \tag{11}$$

Eq. (11) follows from Eq. (10) by differentiation. Conversely, for a connected Lie group, Eq. (11) determines the corresponding covariance condition at the group level.

We will say that the encoding channel $\mathcal{E}$ is *G-covariant*, or *$\mathfrak{g}$-covariant*, when its encoding isometry satisfies Eq. (10), or equivalently Eq. (11). We will often specify covariance at the Lie-algebra level, because the physical representations used below are most naturally described by their infinitesimal generators. This is also the natural language in the analog-simulation setting, where the relevant algebras have the form

$$\mathfrak{su}(d_1) \oplus \mathfrak{su}(d_2) \oplus \cdots .$$

When the representations are clear from context, we simply say that the code is *G-covariant*.

b. Intertwiners. The covariance condition strongly restricts the possible encoding maps. It is precisely the condition that $\mathcal{V}$ is an *intertwiner*. Recall that if (V, ρ_V) and (W, ρ_W) are representations of a group G, then a linear map $A : V \to W$ is a *G-intertwiner* if

$$A \circ \rho_V(g) = \rho_W(g) \circ A, \qquad \forall g \in G. \tag{12}$$

Here G may be finite or continuous. At the Lie-algebra level, for infinitesimal representations $d\rho_V$ and $d\rho_W$, this condition becomes

$$A \circ d\rho_V(X) = d\rho_W(X) \circ A, \qquad \forall X \in \mathfrak{g}.$$

The structure of intertwiners is controlled by Schur's lemma. If V_λ and V_μ are irreducible complex representations of G, then

$$\mathrm{Hom}_G(V_\lambda, V_\mu) = \begin{cases} 0, & \lambda \not\simeq \mu, \\ \mathbb{C} \cdot I_{V_\lambda}, & \lambda \simeq \mu. \end{cases}$$

Thus an intertwiner cannot map an irreducible representation to an inequivalent irreducible representation – it can only map it to an equivalent copy.

More generally, suppose that two finite-dimensional G-representations decompose as

$$W \cong \bigoplus_{\lambda \in \Lambda} V_\lambda \otimes M_\lambda, \qquad V \cong \bigoplus_{\lambda \in \Lambda} V_\lambda \otimes N_\lambda,$$

where Λ is the set of irreducible representations of G, and M_λ and N_λ are multiplicity spaces. Then every intertwiner $A : W \to V$ has the form

$$A \cong \bigoplus_{\lambda \in \Lambda} I_{V_\lambda} \otimes A_\lambda, \qquad A_\lambda \in \mathrm{Hom}(M_\lambda, N_\lambda). \tag{13}$$

Thus the intertwiner acts trivially on the irrep factor V_λ and only acts nontrivially on the corresponding multiplicity spaces.

This observation will be used repeatedly. Besides the encoding isometry itself, the linear extensions of reduced-state maps and the compression map

$$X \longmapsto \mathcal{V}^\dagger X \mathcal{V}$$

are also intertwiners. This is why representation theory fixes much of the structure of the reduced states and compressed local operators appearing below. Throughout the paper, we use the terms covariant map and intertwiner interchangeably: the term covariant is standard in quantum information theory, especially for channels and encodings, while intertwiner is the standard representation-theoretic term.

c. The $SU(d)$ representation setting. The main Lie group considered in this work is $SU(d)$. Since $SU(d)$ is compact, connected, and simply connected, its irreducible representations are in one-to-one correspondence with irreducible representations of $\mathfrak{su}(d)$. We will therefore freely pass between $SU(d)$ and $\mathfrak{su}(d)$ representations. Background on these representation-theoretic facts can be found in Ref. [61].

The physical system consists of n identical subsystems, each carrying the same irreducible representation of $SU(d)$. Thus, the physical Hilbert space is

$$\mathcal{H}_P := (V_{\lambda_0})^{\otimes n}, \tag{14}$$

where V_{λ_0} is a fixed irreducible representation of $SU(d)$. The transversal group action is

$$U_P(g) = \bigotimes_{i=1}^{n} U_{\lambda_0}^{(i)}(g), \qquad g \in SU(d), \tag{15}$$

where $U_{\lambda_0}^{(i)}(g)$ acts as $U_{\lambda_0}(g)$ on the ith tensor factor. At the Lie-algebra level, the derived representation is

$$\pi_P(X) = \sum_{i=1}^{n} \pi_{\lambda_0}^{(i)}(X), \qquad X \in \mathfrak{su}(d), \tag{16}$$

where $\pi_{\lambda_0}^{(i)}(X)$ acts as $\pi_{\lambda_0}(X)$ on the ith tensor factor and as the identity on all other tensor factors. For illustration of Lie group and corresponding Lie algebra actions, see Fig. 1(a).

The logical Hilbert space is the fundamental representation of $SU(d)$:

$$\mathcal{H}_L := V_{\omega_1} \cong \mathbb{C}^d. \tag{17}$$

The notation V_{ω_1} is explained in Appendix S3. We denote the corresponding group action by

$$U_L(g) = u(g), \qquad g \in SU(d),$$

and the corresponding Lie-algebra representation by

$$\pi_L(X) = \bar{X}, \qquad X \in \mathfrak{su}(d).$$

The bar is used only when we want to emphasize that the generator acts on the logical fundamental representation, rather than on one of the physical tensor factors.

d. Multiplicity-vector form of the encoding. Decompose the physical Hilbert space into irreducible $SU(d)$-representations:

$$(V_{\lambda_0})^{\otimes n} \cong \bigoplus_{\lambda \in \Lambda} V_\lambda \otimes M_\lambda. \tag{18}$$

Here $V_\lambda \otimes M_\lambda$ is the isotypic component associated with the irrep V_λ, and M_λ is its multiplicity space [45]. Since the logical space is V_{ω_1}, Eq. (13) implies that the image of any covariant encoding must lie inside the corresponding isotypic component

$$V_{\omega_1} \otimes M_{\omega_1} \subseteq \mathcal{H}_P.$$

Thus a covariant encoding selects one copy of V_{ω_1} inside the physical Hilbert space. Equivalently, it is specified by a *multiplicity vector* $|m_{\omega_1}\rangle \in M_{\omega_1}$:

$$\mathcal{V} : \mathcal{H}_L \cong V_{\omega_1} \longrightarrow V_{\omega_1} \otimes |m_{\omega_1}\rangle \subseteq \mathcal{H}_P; \tag{19}$$

see Fig. 1(b) for schematic illustration. The choice of $|m_{\omega_1}\rangle$ determines the particular covariant code. The same multiplicity-space viewpoint was used in Ref. [60] to reconstruct known quantum error-correcting codes covariant under finite symmetry groups.

This construction is possible only if the fundamental isotypic component is present in the physical representation, that is, only if

$$\dim M_{\omega_1} > 0.$$

For each explicit construction below, we will specify conditions ensuring this nonzero multiplicity.

It is useful to describe the encoding in terms of the Schur transform. For a representation $\mathcal{H}$ of $SU(d)$, the Schur transform is a unitary

$$U_{\text{Schur}} : \mathcal{H} \longrightarrow \bigoplus_{\lambda \in \Lambda} V_\lambda \otimes M_\lambda$$

such that

$$U_{\text{Schur}} U(g) U_{\text{Schur}}^\dagger = \bigoplus_{\lambda \in \Lambda} U_\lambda(g) \otimes I_{M_\lambda}. \tag{20}$$

The Schur transform is not canonical, because it depends on choices of bases in the irreducible spaces V_λ and in the multiplicity spaces M_λ. Its construction in quantum information settings is discussed in Ref. [62].

With this notation, the encoding associated with the multiplicity vector $|m_{\omega_1}\rangle \in M_{\omega_1}$ is

$$\mathcal{V} = U_{\text{Schur}}^\dagger \left(I_{V_{\omega_1}} \otimes |m_{\omega_1}\rangle \right). \tag{21}$$

e. Permutation symmetry of the code space. In addition to the $SU(d)$ action, we will use finite permutation symmetries of the physical subsystems. Let $G_P \subseteq S_n$ act on $\mathcal{H}_P$ by permuting tensor factors:

$$P_g \left(|i_1\rangle \otimes \cdots \otimes |i_n\rangle \right) := \left| i_{g^{-1}(1)} \right\rangle \otimes \cdots \otimes \left| i_{g^{-1}(n)} \right\rangle. \tag{22}$$

This permutation action commutes with the transversal $SU(d)$ action in Eq. (15), because $SU(d)$ acts identically on each tensor factor. Hence G_P preserves the isotypic decomposition (18). By the double-centralizer structure of the commutant, the action of G_P has the form [63]

$$P_g = \bigoplus_{\lambda \in \Lambda} I_{V_\lambda} \otimes R_\lambda(g), \tag{23}$$

where R_λ is a representation of G_P on the multiplicity space M_λ. Thus G_P acts trivially on the $SU(d)$-irrep factor and nontrivially only on the multiplicity space.

The purpose of introducing G_P is to impose an additional symmetry on the code space. After covariance fixes the relevant isotypic component, the remaining freedom is the choice of the multiplicity vector. We choose this vector from the G_P-invariant subspace

$$M_{\omega_1}^{G_P} := \{ |m\rangle \in M_{\omega_1} : R_{\omega_1}(g) |m\rangle = |m\rangle \ \ \forall g \in G_P \}. \tag{24}$$

If $|m\rangle \in M_{\omega_1}^{G_P}$, then the code space

$$V_{\omega_1} \otimes |m\rangle$$

is invariant under G_P, because G_P acts as the identity on V_{ω_1} and preserves $|m\rangle$ in the multiplicity space.

Finally, the same permutation action induces an action on operators by conjugation. For $X \in \mathrm{End}(\mathcal{H}_P)$ and $g \in G_P$, define

$$g \cdot X := P_g X P_g^\dagger, \tag{25}$$

where P_g is the permutation operator from Eq. (22).

IV. $SU(2)$-COVARIANT APPROXIMATE QUANTUM ERROR CORRECTION CODES

We begin the explicit construction with the familiar case of $SU(2)$. The logical system is a spin-1/2 degree of freedom, while the physical system consists of many spin-j subsystems. Covariance fixes the symmetry sector in which the code can live, but it still leaves a multiplicity-space freedom. We use this remaining freedom to distribute the logical spin uniformly over the physical sites. For single-site erasure it means that the environment receives a one-site reduced state, and the code is good when this reduced state is nearly independent of the logical input.

a. Setup and reduced-state strategy. Let V_j denote the irreducible spin-j representation of $SU(2)$. We take the logical Hilbert space (17) to be

$$\mathcal{H}_L = V_{1/2} \cong \mathbb{C}^2,$$

and the physical Hilbert space (14) to be

$$\mathcal{H}_P = (V_j)^{\otimes n},$$

where n is odd and $j \in \frac{1}{2} + \mathbb{Z}$ is half-integer. These conditions are exactly those ensuring that the spin-1/2 representation appears in $(V_j)^{\otimes n}$, or equivalently that the multiplicity space $M_{V_{1/2}}$ is nonzero:

$$\dim M_{V_{1/2}} > 0 \quad \iff \quad n \equiv 1 \pmod 2 \quad \text{and} \quad j \in \frac{1}{2} + \mathbb{Z}.$$

We use the standard spin-operator notation for the physical representation (16) of the generators of $\mathfrak{su}(2)$,

$$\pi_j^{(k)}\left(\frac{i\sigma_a}{2}\right) \equiv J_a^{(k)}, \qquad a = x, y, z,$$

so that the transversal physical action is

$$\frac{i\sigma_a}{2} \longmapsto iJ_a := \sum_{k=1}^{n} iJ_a^{(k)}, \qquad a = x, y, z. \tag{26}$$

This $SU(2)$ setting is useful both as a physically transparent example and as a complete illustration of the general method. The main object is the one-site reduced state $\rho^{(i)}(\rho_L)$ defined in Eq. (9). For flagged single-qudit erasure, this is precisely the state received by the environment, as follows from the complementary channel in Eq. (7). Thus, the task is to choose the code so that $\rho^{(i)}(\rho_L)$ approaches a fixed state, independent of ρ_L, as n grows. A detailed derivation of the final expression (32) is given in Proposition S2.1, here we spell out the representation-theoretic mechanism.

b. Reduced-state structure from covariance. The form of $\rho^{(i)}(\rho_L)$ is strongly constrained by covariance. It satisfies

$$\rho^{(i)}\big(u(g)\rho_L u(g)^\dagger\big) = U_j^{(i)}(g)\rho^{(i)}(\rho_L)\big(U_j^{(i)}(g)\big)^\dagger. \tag{27}$$

Equivalently, after extending $\rho^{(i)}$ linearly from density matrices to all operators on the logical space, the resulting map

$$\mathrm{End}(V_{1/2}) \xrightarrow{\Phi^{(i)}} \mathrm{End}(V_j)$$

is an $SU(2)$-intertwiner (12), where both operator spaces carry the adjoint action. We can therefore determine its possible form from representation theory. On the logical space,

$$\mathrm{End}(V_{1/2}) \cong V_{1/2} \otimes V_{1/2}^* \cong V_0 \oplus V_1,$$

where V_0 is the trivial representation, spanned by the identity, and V_1 is the adjoint representation, spanned by the Pauli generators. On a physical spin-j site,

$$\mathrm{End}(V_j) \cong V_j \otimes V_j^* \cong \bigoplus_{\ell=0}^{2j} V_\ell.$$

Thus Eq. (13) implies that $\Phi^{(i)}$ maps the trivial component of $\mathrm{End}(V_{1/2})$ to the unique trivial component of $\mathrm{End}(V_j)$, and maps the adjoint component of $\mathrm{End}(V_{1/2})$ to the unique adjoint component of $\mathrm{End}(V_j)$.

Writing the logical state in Bloch form,

$$\rho_L = \frac{1}{2}(I + \mathbf{r} \cdot \boldsymbol{\sigma}),$$

the identity term belongs to the trivial representation and therefore maps to a multiple of the identity on V_j. Normalization fixes this multiple to be $1/(2j+1)$. The traceless term belongs to the adjoint representation, and hence its image must be proportional to the unique adjoint component on $\mathrm{End}(V_j)$, spanned by the physical spin operators $J_a^{(i)}$. Therefore the reduced state has the form

$$\rho^{(i)}(\rho_L) = \frac{I_{2j+1}}{2j+1} + \beta^{(i)} \sum_{a=x,y,z} r_a J_a^{(i)}. \tag{28}$$

At this stage the code space has not yet been specified; all dependence on the multiplicity vector (19) is contained in the scalar coefficients $\beta^{(i)}$.

To determine these coefficients, we first compress local spin operators to the code space. Since the encoding is covariant, the total physical spin acts as the logical spin,

$$\mathcal{V}^\dagger \sum_{i=1}^{n} J_a^{(i)} \mathcal{V} = \frac{\sigma_a}{2}. \tag{29}$$

Moreover, the compression map $X \mapsto \mathcal{V}^\dagger X \mathcal{V}$ is again an $SU(2)$-intertwiner $\mathrm{End}(V_j) \to \mathrm{End}(V_{1/2})$. Since the operators $J_a^{(i)}$ span an adjoint component on the ith site, Eq. (13) implies that their compression must be proportional to the unique adjoint component on the logical spin-1/2 space. Hence

$$\mathcal{V}^\dagger J_a^{(i)} \mathcal{V} = \alpha^{(i)} \frac{\sigma_a}{2}, \tag{30}$$

for real scalars $\alpha^{(i)}$ satisfying $\sum_i \alpha^{(i)} = 1$, where the latter identity follows from Eq. (29). Comparing Hilbert–Schmidt inner products in Eqs. (28) and (30) gives

$$\beta^{(i)} = \frac{3\alpha^{(i)}}{2j(j+1)(2j+1)}. \tag{31}$$

c. Cyclic invariance. It remains to choose the multiplicity vector so that the coefficients $\alpha^{(i)}$ are uniform. We do this by choosing $|m\rangle$ to be an eigenvector of the cyclic shift $s \in C_n \subseteq S_n$. Equivalently, the code space is invariant under cyclic permutations of the physical qudits. If P_s denotes the corresponding permutation operator, then $P_s J_a^{(i)} P_s^\dagger = J_a^{(i+1)}$, while on the code space $P_s \mathcal{V} = e^{i\phi}\mathcal{V}$. Therefore

$$\mathcal{V}^\dagger J_a^{(i+1)} \mathcal{V} = \mathcal{V}^\dagger P_s J_a^{(i)} P_s^\dagger \mathcal{V} = \mathcal{V}^\dagger J_a^{(i)} \mathcal{V}.$$

All coefficients $\alpha^{(i)}$ are equal. Since they sum to one, $\alpha^{(i)} = 1/n$ for every site.

For the cyclically invariant $SU(2)$-covariant code, the one-site reduced state is therefore

$$\rho^{(i)}(\rho_L) = \frac{I_{2j+1}}{2j+1} + \frac{3}{2nj(j+1)(2j+1)} \sum_{a=x,y,z} r_a J_a^{(i)}, \tag{32}$$

where $i = 1, \dots, n$ and $\rho_L = \frac{1}{2}(I + \mathbf{r}\cdot\boldsymbol{\sigma})$. This expression makes the error-correction mechanism transparent, because the logical-state-dependent part of the erased site's density matrix is suppressed by $1/n$, so the environment's state approaches the maximally mixed state as n grows.

d. Single-site erasure bound. We now translate this reduced-state statement into an AQEC bound. Since all one-site reduced states are identical up to the natural identification of the physical sites, the complementary channel for flagged single-site erasure becomes

$$\widehat{\mathcal{N}\circ\mathcal{E}}(\rho_L) = \sum_i p_i \, |i\rangle\langle i|_{F_E} \otimes \rho^{(i)}(\rho_L) \cong \omega_{\text{flag}} \otimes \rho^{(1)}(\rho_L), \tag{33}$$

where

$$\omega_{\text{flag}} := \sum_i p_i \, |i\rangle\langle i|_{F_E}.$$

We compare this channel with the constant channel

$$\Lambda_0(\rho) := \mathrm{Tr}(\rho)\, \omega_{\text{flag}} \otimes \frac{I_{2j+1}}{2j+1}.$$

The explicit fidelity calculation, given in Proposition S2.3, yields

$$F\left(\widehat{\mathcal{N}\circ\mathcal{E}}, \Lambda_0\right) = 1 - \frac{9}{32n^2 j(j+1)} + O_j(n^{-3}). \tag{34}$$

Together with the complementary-channel characterization of AQEC, this yields the following result.

Theorem 2. *Let $\mathcal{H}_L = V_{1/2}$ and $\mathcal{H}_P = (V_j)^{\otimes n}$, with n odd and $j \in \frac{1}{2} + \mathbb{Z}$. Equip $\mathcal{H}_L$ with the fundamental spin-1/2 representation of $\mathfrak{su}(2)$ and $\mathcal{H}_P$ with the transversal spin-j representation*

$$\frac{i\sigma_a}{2} \longmapsto iJ_a \equiv \sum_{k=1}^{n} iJ_a^{(k)}, \qquad a = x, y, z. \tag{35}$$

Let $\mathcal{E}$ be an $\mathfrak{su}(2)$-covariant encoding whose code space is invariant under cyclic permutations of the physical qudits. Then, for the flagged single-qudit erasure channel

$$\mathcal{N}(\sigma) = \sum_{i=1}^{n} p_i \, |i\rangle\langle i|_F \otimes |e\rangle\langle e|_{A_i} \otimes \mathrm{Tr}_i(\sigma),$$

there exists a recovery channel $\mathcal{R}$ such that

$$d(\mathcal{R}\mathcal{N}\mathcal{E}, \mathrm{id}) \le \frac{3}{4\sqrt{2j(j+1)}} \frac{1}{n} + O_j(n^{-2}).$$

In particular, the code is $O\left(1/(\sqrt{j(j+1)}\, n)\right)$-correctable against flagged single-qudit erasure.

Proof. By Eq. (34) and the definition of the channel distance in Eq. (1),

$$d\left(\widehat{\mathcal{N}\circ\mathcal{E}}, \Lambda_0\right) = \frac{3}{4\sqrt{2j(j+1)}} \frac{1}{n} + O_j(n^{-2}).$$

The complementary-channel theorem, Eq. (5), gives

$$\min_{\mathcal{R}} d(\mathcal{R}\mathcal{N}\mathcal{E}, \mathrm{id}) = \min_{\Lambda\,\text{constant}} d\left(\widehat{\mathcal{N}\circ\mathcal{E}}, \Lambda\right) \le d\left(\widehat{\mathcal{N}\circ\mathcal{E}}, \Lambda_0\right),$$

and the stated bound follows. □

Known lower bounds for covariant codes under erasure noise [42] give

$$\min_{\mathcal{R}} d(\mathcal{R}\mathcal{N}\mathcal{E}, \mathrm{id}) \ge \frac{1}{4\sqrt{2}\, j} \frac{1}{n}.$$

Thus the construction is optimal in its scaling with n and j, up to constant factors.

V. $SU(d)$-COVARIANT APPROXIMATE QUANTUM ERROR CORRECTION CODES

We now extend the construction to $SU(d)$. The logical system carries the fundamental representation, while each physical subsystem carries an antisymmetric tensor-power representation. As in the $SU(2)$ case, covariance fixes the symmetry sector in which the code can live, but leaves a freedom in the corresponding multiplicity space. We use this freedom to choose a code space that is invariant under suitable permutations of the physical subsystems. For single-qudit flagged erasure, cyclic permutation symmetry is sufficient, since it spreads the logical information uniformly over the physical sites, so that each erased-site reduced state becomes nearly independent of the logical input.

a. Single-site physical representation. We first define the single-site physical representation. For $1 \le r \le d-1$, let

$$V_{\omega_r} := \bigwedge^r \mathbb{C}^d. \tag{36}$$

This space has basis vectors

$$e_I := e_{i_1} \wedge \cdots \wedge e_{i_r}, \qquad 1 \le i_1 < \cdots < i_r \le d,$$

where $\wedge$ denotes the wedge product. Its dimension is

$$\dim V_{\omega_r} = \binom{d}{r}.$$

The action of $X \in \mathfrak{su}(d)$ on V_{ω_r} is defined by

$$\pi_{\omega_r}(X)\,(v_1 \wedge \cdots \wedge v_r) := \sum_{m=1}^{r} v_1 \wedge \cdots \wedge X v_m \wedge \cdots \wedge v_r. \tag{37}$$

This is an irreducible representation; see Ref. [64]. The notation V_{ω_r} is explained in Appendix S3.

We use the standard generators of the fundamental representation of $\mathfrak{su}(d)$. Let $t_a{}_{a=1}^{d^2-1}$ be traceless Hermitian $d \times d$ matrices satisfying

$$t_a t_b = \frac{1}{2d}\delta_{ab} I_d + \frac{1}{2}\sum_{c=1}^{d^2-1} \left(i f_{abc} + d_{abc}\right) t_c, \tag{38}$$

where f_{abc} are completely antisymmetric structure constants and d_{abc} are completely symmetric coefficients. Further properties of these generators are collected in Appendix S3.

We take the logical Hilbert space (17) to be the fundamental representation

$$\mathcal{H}_L := V_{\omega_1} \cong \mathbb{C}^d,$$

and the physical Hilbert space (14) to be

$$\mathcal{H}_P := (V_{\omega_r})^{\otimes n},$$

where $r \neq 0, d$ and $nr \equiv 1 \pmod d$. We denote the action of the generator t_a on the jth physical subsystem by

$$\pi^{(j)}_{\omega_r}(t_a) \equiv T_a^{(j)}, \qquad a = 1, \ldots, d^2-1. \tag{39}$$

Thus the transversal physical representation (16) is specified on generators by

$$t_a \longmapsto T_a := \sum_{i=1}^{n} T_a^{(i)}, \qquad a = 1, \ldots, d^2-1.$$

In this section the representation V_{ω_r} is fixed, so we omit the subscript ω_r from the local generators. In later sections and in the main theorems, we will often specify the physical representation by giving the images of the Lie-algebra generators.

The condition $nr \equiv 1 \pmod d$ ensures that $\mathcal{H}_P$ contains a copy of the fundamental representation. Equivalently, the V_{ω_1}-isotypic component is nonzero, so it can contain the image of an $SU(d)$-covariant encoding of the logical space. Lemma S3.5 proves that this condition is sufficient for $(V_{\omega_r})^{\otimes n}$ to contain V_{ω_1}. It is not necessary in general. For example, the $SU(2)$ construction in Section IV used arbitrary half-integer spin-j irreducible representations as physical subsystems.

b. One-site reduced state. The main object is again the one-site reduced state $\rho^{(i)}(\rho_L)$ defined in Eq. (9). A detailed derivation of its final form is given in Proposition S3.7. Here we describe the representation-theoretic structure of the calculation.

Write the logical state in the generalized Bloch form

$$\rho_L = \frac{I}{d} + \left(\rho_L - \frac{I}{d}\right) = \frac{I}{d} + \sum_a r_a t_a. \tag{40}$$

The reduced-state map extends linearly to a map

$$\mathrm{End}(V_{\omega_1}) \xrightarrow{\Phi^{(i)}} \mathrm{End}(V_{\omega_r}).$$

By covariance, this map is an $SU(d)$-intertwiner (12), where both operator spaces carry the adjoint action. As shown in Lemma S3.6, $\mathrm{End}(V_{\omega_r})$ contains unique copies of both the trivial representation and the adjoint representation. Therefore, using the intertwiner decomposition (13), the identity component of $\mathrm{End}(V_{\omega_1})$ must map to the identity component of $\mathrm{End}(V_{\omega_r})$, while the adjoint component must map to the adjoint component. Hence,

$$\begin{aligned} \Phi^{(i)}\left(\frac{I_d}{d}\right) &= \frac{I_{V_{\omega_r}}}{\dim V_{\omega_r}}, \\ \Phi^{(i)}\left(\sum_a r_a t_a\right) &= \beta^{(i)} \sum_a r_a T_a^{(i)}. \end{aligned} \tag{41}$$

Thus the reduced state has the form

$$\rho^{(i)}(\rho_L) = \frac{I_{V_{\omega_r}}}{\dim V_{\omega_r}} + \beta^{(i)} \sum_a r_a T_a^{(i)}. \tag{42}$$

The coefficients $\beta^{(i)}$ are determined by the compression of local generators to the code space. Since the encoding is covariant, the total physical action restricts to the logical fundamental action. For each local generator, the compression map $X \mapsto \mathcal{V}^\dagger X \mathcal{V}$ is again an $SU(d)$-intertwiner $\mathrm{End}(V_{\omega_r}) \to \mathrm{End}(V_{\omega_1})$, and therefore

$$\mathcal{V}^\dagger T_a^{(i)} \mathcal{V} = \alpha^{(i)} \bar{t}_a,$$

where

$$\sum_i \alpha^{(i)} = 1,$$

and $\bar{t}_a$ denotes the fundamental generator acting on the logical space. The relation between $\alpha^{(i)}$ and $\beta^{(i)}$ is obtained by comparing Hilbert–Schmidt inner products, exactly as in the $SU(2)$ case, giving

$$\beta^{(i)} = \frac{\alpha^{(i)}}{2\kappa_{\omega_r}}, \qquad \kappa_{\omega_r} = \frac{1}{2}\binom{d-2}{r-1}.$$

It remains to choose the multiplicity vector. We choose $|m\rangle$ to be an eigenvector of the cyclic shift operator. Equivalently, the code space $V_{\omega_1} \otimes |m\rangle$ is invariant under cyclic permutations of the physical qudits. The same argument as in Section IV then implies that all coefficients $\alpha^{(i)}$ are equal. Since they sum to one,

$$\alpha^{(i)} = \frac{1}{n} \qquad \text{for every } i.$$

For this cyclically invariant $SU(d)$-covariant code, the one-site reduced state becomes

$$\rho^{(i)}(\rho_L) = \frac{I_{V_{\omega_r}}}{\dim V_{\omega_r}} + \frac{1}{2\kappa_{\omega_r} n} \sum_a r_a T_a^{(i)}.$$

The logical-state-dependent part is suppressed by $1/n$, so the erased-site state approaches the maximally mixed state as n grows; see Fig. 1(c) for a schematic illustration.

c. Single-site erasure bound. We now translate this reduced-state statement into an AQEC bound. Since the one-site reduced states are identical up to the natural identification of physical sites, the complementary channel (7) for flagged single-qudit erasure is

$$\widehat{\mathcal{N} \circ \mathcal{E}}(\rho_L) \cong \omega_{\text{flag}} \otimes \left(\frac{I_{V_{\omega_r}}}{\dim V_{\omega_r}} + \frac{1}{2\kappa_{\omega_r} n} \sum_a r_a T_a^{(1)} \right),$$

where

$$\omega_{\text{flag}} := \sum_i p_i \, |i\rangle \langle i|_{F_E} .$$

We compare this channel with the constant channel (4)

$$\Lambda_0(\rho) := \operatorname{Tr}(\rho)\, \omega_{\text{flag}} \otimes \frac{I_{V_{\omega_r}}}{\dim V_{\omega_r}}.$$

The explicit fidelity calculation, given in Proposition S3.8, yields

$$F\left(\widehat{\mathcal{N} \circ \mathcal{E}}, \Lambda_0\right) = 1 - \frac{(d-1)^2(d+1)}{8r(d-r)} \frac{1}{n^2} + O_d(n^{-3}). \tag{43}$$

Together with the complementary-channel characterization of AQEC, this gives the following result.

Theorem 3. *Let $\mathcal{H}_L = V_{\omega_1}$ and $\mathcal{H}_P = (V_{\omega_r})^{\otimes n}$, with $r \neq 0, d$ and $nr \equiv 1 \pmod d$. Equip $\mathcal{H}_L$ with the fundamental representation of $\mathfrak{su}(d)$ and $\mathcal{H}_P$ with the transversal representation*

$$t_a \longmapsto T_a \equiv \sum_{i=1}^{n} T_a^{(i)}, \qquad a = 1, \dots, d^2-1, \tag{44}$$

where $T_a^{(i)}$ is the representation of t_a on the ith physical space V_{ω_r}, acting as in Eq. (37). Let $\mathcal{E}$ be an $\mathfrak{su}(d)$-covariant encoding whose code space is invariant under cyclic permutations of the physical qudits. Then, for the flagged single-qudit erasure channel

$$\mathcal{N}(\sigma) = \sum_{i=1}^{n} p_i \, |i\rangle \langle i|_F \otimes |e\rangle \langle e|_{A_i} \otimes \operatorname{Tr}_i(\sigma),$$

there exists a recovery channel $\mathcal{R}$ such that

$$d(\mathcal{R}\mathcal{N}\mathcal{E}, \mathrm{id}) \leq \frac{d-1}{2\sqrt{2}} \sqrt{\frac{d+1}{r(d-r)}} \frac{1}{n} + O_d(n^{-2}).$$

In particular, the code is

$$O\left((d-1)\sqrt{\frac{d+1}{r(d-r)}}\frac{1}{n}\right)$$

-correctable against flagged single-qudit erasure.

Proof. By Eq. (43) and the definition of the channel distance in Eq. (1),

$$d\left(\widehat{\mathcal{N} \circ \mathcal{E}}, \Lambda_0\right) = \frac{d-1}{2\sqrt{2}} \sqrt{\frac{d+1}{r(d-r)}} \frac{1}{n} + O_d(n^{-2}).$$

The complementary-channel theorem, Eq. (5), gives

$$\min_{\mathcal{R}} d(\mathcal{R}\mathcal{N}\mathcal{E}, \mathrm{id}) = \min_{\Lambda, \text{constant}} d\left(\widehat{\mathcal{N} \circ \mathcal{E}}, \Lambda\right) \leq d\left(\widehat{\mathcal{N} \circ \mathcal{E}}, \Lambda_0\right),$$

and the stated bound follows. □

For $r = 1$, the upper bound coincides with the scaling obtained in Ref. [45] for a randomized code construction. The lower bound of Ref. [42] gives

$$\min_{\mathcal{R}} d(\mathcal{R}\mathcal{N}\mathcal{E}, \mathrm{id}) \geq \frac{1}{2n}.$$

Thus the construction is optimal in its scaling with n, up to a constant factor depending on d and r.

A. Covariant Decoder

We now discuss an explicit decoder for the covariant codes constructed above. In approximate quantum error correction, the recovery map is not fixed uniquely by the code and the noise model. Instead, one has to choose a recovery channel that gives a small error with respect to the chosen distance measure. Determining the optimal recovery, including the sharp constants in its dependence on the physical parameters, can be difficult. We therefore use a natural recovery map that is known to be near-optimal.

a. Petz recovery map. We use the transpose channel, also called the Petz recovery map. For a channel $\mathcal{N}$ and a reference state ω, the Petz map is

$$\mathcal{P}_{\omega,\mathcal{N}}(\sigma) = \omega^{1/2} \mathcal{N}^* \left(\mathcal{N}(\omega)^{-1/2} \sigma \mathcal{N}(\omega)^{-1/2} \right) \omega^{1/2},$$

where $\mathcal{N}^*$ is the adjoint of $\mathcal{N}$ with respect to the Hilbert–Schmidt inner product, and the inverse is taken on the support of $\mathcal{N}(\omega)$.

b. Near-optimality. The analysis uses two main facts. First, for the reference state $\omega = I_d/d$, the Petz-recovered logical channel

$$\Phi := \mathcal{R}_{\text{Petz}} \circ \mathcal{N} \circ \mathcal{E}$$

is $SU(d)$-covariant. For related uses of channel covariance, see Refs. [43, 48]. Since the logical representation is irreducible, Schur's lemma implies that Φ is a depolarizing channel:

$$\Phi(\rho) = \lambda_{\text{Petz}}\rho + (1 - \lambda_{\text{Petz}}) \frac{I_d}{d} \operatorname{Tr} \rho.$$

The depolarizing parameter is

$$\lambda_{\text{Petz}} = \frac{2}{d^2 - 1} \sum_{i=1}^{n} p_i \sum_{a=1}^{d^2-1} \operatorname{Tr} \left[S_i^{-1/2} \mathcal{C}_i(t_a) S_i^{-1/2} \mathcal{C}_i(t_a) \right], \tag{45}$$

where

$$\mathcal{C}_i(\rho_L) := \operatorname{Tr}_i \left(\mathcal{V} \rho_L \mathcal{V}^\dagger \right), \qquad S_i := \operatorname{Tr}_i \left(\mathcal{V}\mathcal{V}^\dagger \right).$$

The corresponding entanglement fidelity is

$$F(\Phi, \text{id}) = \sqrt{\lambda_{\text{Petz}} + \frac{1 - \lambda_{\text{Petz}}}{d^2}}.$$

Thus, determining the Petz recovery error with sharp constants is equivalent to determining the asymptotics of λ_{Petz}.

The exact expression in Eq. (45) is difficult to analyze directly. It requires control of the large-n behavior of the operators $S_i^{-1/2}$ and $\mathcal{C}_i(t_a)$ on the $(n-1)$-site space. These operators have support in representation sectors of $(V_{\omega_r})^{\otimes(n-1)}$ where the adjoint representation can appear with large multiplicity. This multiplicity structure makes a direct asymptotic calculation substantially more involved than the one-site reduced-state calculation. Similar difficulties also appear in the analysis of multi-qudit erasures below.

For this reason, we use the near-optimality of the transpose channel established in Ref. [57]. Combined with the bound in Theorem 3, this shows that the Petz recovery achieves the same $1/n$ scaling of the recovery error, up to a constant factor. The explicit form of the Petz decoder and its asymptotic performance are stated in Theorem S3.10:

Theorem 4. *For the encoding $\mathcal{E}$ and the flagged single-qudit erasure channel $\mathcal{N}$ defined in Theorem 3, the Petz recovery map with reference state $\omega = I_d/d$ has the form*

$$\mathcal{R}_{\text{Petz}}(X) = \sum_{i:,p_i>0} \mathcal{V}^\dagger \left(S_i^{-1/2} \langle i, e|X|i, e\rangle_{F,A_i} S_i^{-1/2} \otimes I_{A_i} \right) \mathcal{V},$$

where $\mathcal{V}$ is the encoding isometry and

$$S_i := \operatorname{Tr}_i \left(\mathcal{V}\mathcal{V}^\dagger \right).$$

Moreover, this recovery is near-optimal in the sense that

$$d \left(\mathcal{R}_{\text{Petz}} \circ \mathcal{N} \circ \mathcal{E}, \text{id} \right) = O_d(n^{-1}).$$

This result gives the asymptotic scaling in n, but not the sharp constants depending on the representation-theoretic parameters. We therefore do not claim constant-level optimality of the Petz recovery map in this setting. A more detailed analysis of Eq. (45), including its dependence on the relevant multiplicity spaces, is left for future work.

The implementation of the decoder is closely related to the Schur transform in Eq. (20). The Schur transform appears explicitly in the encoding isometry $\mathcal{V}$ in Eq. (21), and it also enters the Petz map through the operators S_i.

B. Flagged two- and three-qudit erasure

We now consider flagged erasure errors affecting more than one physical site. In this subsection each physical subsystem carries the fundamental representation of $SU(d)$, so that

$$\mathcal{H}_P = (V_{\omega_1})^{\otimes n}.$$

The transversal physical representation is specified on generators by

$$t_a \longmapsto \sum_{i=1}^{n} t_a^{(i)}, \qquad a = 1, \ldots, d^2 - 1,$$

where $t_a^{(i)}$ denotes the fundamental action of t_a on the ith physical space V_{ω_1}.

a. Why higher transitivity is needed. For single-site erasure, the main object was the one-site reduced state. For multi-site erasure, the same role is played by the reduced state on the erased set of sites. Thus, for k-site erasure, the relevant states are

$$\rho^{(i_1,\ldots,i_k)}(\rho_L).$$

As before, these reduced-state maps are covariant. Hence their possible form is constrained by representation theory, while the choice of code space, equivalently the choice of multiplicity vector, enters through the coefficients in the resulting expansion.

The main new issue is that cyclic symmetry is no longer sufficient. In the one-site case, cyclic invariance forced

$$\rho^{(i)}(\rho_L) \cong \rho^{(j)}(\rho_L) \qquad \text{for all } i, j,$$

and this made the logical-state-dependent part of each one-site reduced state scale as $1/n$. For k-site erasure, the analogous condition is that all k-site reduced states be equal up to the canonical identification of tensor factors:

$$\rho^{(i_1,\ldots,i_k)}(\rho_L) \cong \rho^{(j_1,\ldots,j_k)}(\rho_L)$$

for all ordered k-tuples of distinct physical indices. This requires a larger permutation symmetry than the cyclic group.

We therefore choose the multiplicity vector $|m\rangle \in M_{\omega_1}$ to be invariant under a subgroup $G_P \subseteq S_n$ acting on the physical systems as in Eq. (22). The subgroup has to satisfy two requirements. First, it must be sufficiently transitive to identify all relevant reduced states. Second, it must be small enough that the invariant subspace $M_{\omega_1}^{G_P}$ in Eq. (24) is nonzero.

The first requirement is met if G_P is k-transitive. This means that for any two ordered k-tuples of distinct indices $(i_1,\dots,i_k)$ and $(j_1,\dots,j_k)$, there exists $g \in G_P$ such that

$$g(i_1,\dots,i_k) = (j_1,\dots,j_k).$$

If the code space is invariant under this action, then $P_g \mathcal{V}\rho_L \mathcal{V}^\dagger P_g^\dagger = \mathcal{V}\rho_L\mathcal{V}^\dagger$, and the induced action on local operators (25) gives

$$\rho^{(j_1,\dots,j_k)}(\rho_L) = P_g \rho^{(i_1,\dots,i_k)}(\rho_L) P_g^\dagger.$$

b. Permutation groups and invariant multiplicities. The full symmetric group S_n would enforce this condition for every k, but it is too restrictive for the present construction. The multiplicity space M_{ω_1} does not contain a nonzero S_n-invariant subspace, since M_{ω_1} is an irreducible S_n-representation; see Lemma S3.11. Therefore, one has to use a proper subgroup of S_n.

For the construction used here, the relevant proper highly transitive subgroups are the affine group

$$\mathrm{AGL}(1,n) := \left\{x \mapsto ax+b : a \in \mathbb{F}_n^\times, ; b \in \mathbb{F}_n\right\} \tag{46}$$

and the projective linear group

$$\mathrm{PGL}(2,n-1) := \mathrm{GL}\left(2,\mathbb{F}_{n-1}\right)/\mathbb{F}_{n-1}^\times. \tag{47}$$

The group $\mathrm{AGL}(1,n)$ is 2-transitive on $\mathbb{F}_n$, while $\mathrm{PGL}(2,n-1)$ is 3-transitive on the projective line

$$\mathbb{P}^1(\mathbb{F}_{n-1}) = \mathbb{F}_{n-1} \cup \infty.$$

These groups are the useful proper highly transitive subgroups for our construction. Their use is guided by the classification of finite multiply transitive permutation groups, which relies on the Classification of Finite Simple Groups [65]. In particular, apart from the alternating group A_n, there is no infinite family of proper k-transitive subgroups of S_n for $k > 3$. Beyond S_n and A_n, the only 4-transitive groups are the Mathieu groups $M_{11}, M_{12}, M_{23}, M_{24}$, and the only 5-transitive ones are M_{12} and M_{24}; no other group is 6-transitive. The Mathieu groups occur only at $n \in \{11,12,23,24\}$, so they cannot support a construction with $n \to \infty$. The alternating group A_n is a proper highly transitive subgroup of S_n, but it is almost as large as S_n and is too restrictive for the second requirement, since it does not provide the nonzero invariant multiplicity subspaces needed for our code construction. Thus, for our purposes, the useful choices are $\mathrm{AGL}(1,n)$ for $k = 2$ and $\mathrm{PGL}(2,n-1)$ for $k = 3$. For further background on these permutation groups, see Ref. [66].

The existence of nonzero invariant subspaces

$$M_{\omega_1}^{\mathrm{AGL}(1,n)} \neq 0, \qquad M_{\omega_1}^{\mathrm{PGL}(2,n-1)} \neq 0$$

for sufficiently large n is proved in Lemmas S3.15 and S3.18. Therefore, one can choose multiplicity vectors

$$|v_1\rangle \in M_{\omega_1}^{\mathrm{AGL}(1,n)}, \qquad |v_2\rangle \in M_{\omega_1}^{\mathrm{PGL}(2,n-1)}.$$

The corresponding code spaces

$$V_{\omega_1} \otimes |v_1\rangle, \qquad V_{\omega_1} \otimes |v_2\rangle$$

are invariant under $\mathrm{AGL}(1,n)$ and $\mathrm{PGL}(2,n-1)$, respectively.

We summarize the resulting existence statements. The formal proofs are given in Theorems S3.17 and S3.20.

Theorem 5. *Let $\mathcal{H}_L = V_{\omega_1}$ and $\mathcal{H}_P = (V_{\omega_1})^{\otimes n}$, where $n \equiv 1 \pmod d$ and $n = p^k$ for a prime p and a positive integer k. Equip $\mathcal{H}_L$ with the fundamental representation of $\mathfrak{su}(d)$ and $\mathcal{H}_P$ with the transversal representation*

$$t_a \longmapsto \sum_{i=1}^n t_a^{(i)}, \qquad a = 1,\dots,d^2-1,$$

where $t_a^{(i)}$ denotes the fundamental action of t_a on the ith physical space V_{ω_1}. For sufficiently large n, there exists an $\mathfrak{su}(d)$-covariant encoding $\mathcal{E}$ with respect to these representations whose code space is invariant under the action of $\mathrm{AGL}(1,n)$, defined in Eq. (46).

Theorem 6. *Let $\mathcal{H}_L = V_{\omega_1}$ and $\mathcal{H}_P = (V_{\omega_1})^{\otimes n}$. Assume that $d = p^r$, where p is prime and r is a positive integer, and that $n - 1 = p^k$, where $k \geq r$ is a positive integer. Equip $\mathcal{H}_L$ with the fundamental representation of $\mathfrak{su}(d)$ and $\mathcal{H}_P$ with the transversal representation*

$$t_a \longmapsto \sum_{i=1}^n t_a^{(i)}, \qquad a = 1,\dots,d^2-1,$$

where $t_a^{(i)}$ denotes the fundamental action of t_a on the ith physical space V_{ω_1}. For sufficiently large n, there exists an $\mathfrak{su}(d)$-covariant encoding $\mathcal{E}$ with respect to these representations whose code space is invariant under the action of $\mathrm{PGL}(2,n-1)$, defined in Eq. (47).

These results explain why exact permutation symmetry of this kind lets us treat erasure on at most three sites. This restriction is a limitation of the symmetry-based construction used here, not a no-go statement for more general codes. Requiring all k-site reduced states to coincide is a sufficient condition for approximate error correction, but it is not necessary. Section V C removes this restriction by giving up exact symmetry. A typical, rather than exactly symmetric, code is shown to correct erasure on any fixed number of sites.

c. Three-site reduced states. We now focus on flagged erasure of three physical sites. Since $\mathrm{PGL}(2, n-1)$ is also 2-transitive, the same construction also applies to two-site erasures. If the goal is only to correct two-site erasures, the weaker $\mathrm{AGL}(1, n)$ symmetry already suffices. The two-site case is discussed separately in Section VI C.

The three-site reduced-state calculation is more involved than the one-site calculation. The linear extension of the reduced-state map $\rho^{(ijk)}$ is still an intertwiner, but the relevant operator space now contains several copies of the adjoint representation. Covariance therefore no longer fixes the reduced state up to a single scalar coefficient, unlike in the one-site case (41).

The detailed derivation is given in Section S3.2. Here we summarize the main steps. First, Lemmas C.30 and C.31 decompose the relevant trivial and adjoint components in a basis of few-body operators. Next, Proposition C.28 determines the compression of these few-body operators to the code space. For distinct physical sites i, j, k, one obtains

$$\mathcal{V}^\dagger t_a^{(i)} \mathcal{V} = \frac{1}{n}\bar{t}_a, \quad \mathcal{V}^\dagger t_{a_1}^{(i)} t_{a_2}^{(j)} \mathcal{V} = -\frac{1}{2dn}\delta_{a_1 a_2} I_L,$$
$$\mathcal{V}^\dagger t_{a_1}^{(i)} t_{a_2}^{(j)} t_{a_3}^{(k)} \mathcal{V} = \frac{1}{2dn(n-2)} d_{a_1 a_2 a_3} I_L -$$
$$-\frac{1}{2dn(n-2)}\left(\delta_{a_1 a_2}\bar{t}_{a_3} + \delta_{a_1 a_3}\bar{t}_{a_2} + \delta_{a_2 a_3}\bar{t}_{a_1}\right),$$

where $\bar{t}_a$ denotes the fundamental generator acting on the logical space. Finally, Proposition C.34 combines these compression identities with the relevant Hilbert–Schmidt inner-product identities and gives the three-site reduced state

$$\rho^{(ijk)}(\rho_L) = \tau^{(ijk)} + \Delta^{(ijk)}(\rho_L). \tag{48}$$

Here $\tau^{(ijk)}$ is independent of the logical input, while $\Delta^{(ijk)}(\rho_L)$ contains all dependence on ρ_L. Explicitly,

$$\begin{aligned}\tau^{(ijk)} &= \frac{1}{d^3}\mathbf{1} - \frac{2}{d^2 n}\sum_{m=1}^{d^2-1}\left(t_m^{(i)} t_m^{(j)} + t_m^{(i)} t_m^{(k)} + t_m^{(j)} t_m^{(k)}\right) \\ &\quad + \frac{4}{dn(n-2)}\sum_{m,l,p=1}^{d^2-1} d_{mlp}\, t_m^{(i)} t_l^{(j)} t_p^{(k)},\end{aligned} \tag{49}$$

$$\begin{aligned}\Delta^{(ijk)}(\rho_L) &= \sum_b r_b \Bigg[\frac{1}{nd^2}\left(t_b^{(i)} + t_b^{(j)} + t_b^{(k)}\right) \\ &\quad - \frac{2}{dn(n-2)}\sum_{m=1}^{d^2-1}\left(t_b^{(i)} t_m^{(j)} t_m^{(k)} + t_m^{(i)} t_b^{(j)} t_m^{(k)} + t_m^{(i)} t_m^{(j)} t_b^{(k)}\right)\Bigg].\end{aligned} \tag{50}$$

d. Three-site erasure bound. After identifying all triples of physical sites using the 3-transitive symmetry, the complementary channel (8) takes the form

$$\widehat{\mathcal{N}\circ\mathcal{E}}(\rho_L) \cong \omega_{\text{flag}} \otimes \rho^{(i_0 j_0 k_0)}(\rho_L),$$

where i_0, j_0, k_0 are fixed distinct indices and

$$\omega_{\text{flag}} = \sum_{i<j<k} p_{ijk} \left|ijk\right\rangle\left\langle ijk\right|_{F_E}.$$

The appropriate constant channel is obtained by keeping the input-independent part of the three-site reduced state:

$$\Lambda_0(\rho_L) = \mathrm{Tr}(\rho_L)\, \omega_{\text{flag}} \otimes \tau^{(i_0 j_0 k_0)}.$$

Unlike in the one-site case, this state is not simply maximally mixed; it contains nontrivial few-body terms.

The entanglement fidelity is computed explicitly in Proposition S3.34:

$$F\left(\widehat{\mathcal{N}\circ\mathcal{E}}, \Lambda_0\right) = 1 - \frac{3(d^2-1)}{8n^2} + O_d(n^{-3}). \tag{51}$$

Together with the complementary-channel characterization of AQEC, this gives the following result.

Theorem 7. *Let $\mathcal{E}$ be the $\mathfrak{su}(d)$-covariant encoding from Theorem 6. Then, for the flagged three-site erasure channel*

$$\mathcal{N}(\sigma) = \sum_{1\le i<j<k\le n} p_{ijk} \left|ijk\right\rangle\left\langle ijk\right|_F \otimes \left|e\right\rangle\left\langle e\right|_{ijk} \otimes \mathrm{Tr}_{ijk}(\sigma),$$

there exists a recovery channel $\mathcal{R}$ such that

$$d(\mathcal{R}\mathcal{N}\mathcal{E}, \mathrm{id}) \le \frac{\sqrt{3(d^2-1)}}{2\sqrt{2}}\frac{1}{n} + O_d(n^{-2}).$$

In particular, the code is $O(\sqrt{d^2-1}/n)$-correctable against flagged erasure on three sites.

Proof. By Eq. (51) and the definition of the channel distance in Eq. (1),

$$d\left(\widehat{\mathcal{N}\circ\mathcal{E}},\Lambda_0\right)=\frac{\sqrt{3(d^2-1)}}{2\sqrt{2}}\frac{1}{n}+O_d(n^{-2}).$$

The complementary-channel theorem, Eq. (5), gives

$$\min_{\mathcal{R}} d(\mathcal{RNE},\mathrm{id})=\min_{\Lambda,\text{constant}} d\left(\widehat{\mathcal{N}\circ\mathcal{E}},\Lambda\right)\le d\left(\widehat{\mathcal{N}\circ\mathcal{E}},\Lambda_0\right),$$

and the stated bound follows. □

C. Flagged erasure on an arbitrary fixed number of qudits

We now extend the erasure analysis to an arbitrary fixed number k of erased sites, completing the discussion of Section V B. As before, $\mathcal{H}_L=V_{\omega_1}$ and $\mathcal{H}_P=(V_{\omega_1})^{\otimes n}$, with the transversal representation

$$t_a\longmapsto\sum_{i=1}^{n}t_a^{(i)},\qquad a=1,\dots,d^2-1.$$

a. Why exact symmetry fails for general k. For $k\le 3$, Section V B chose the multiplicity vector invariant under a subgroup of S_n that is exactly k-transitive, so that all k-site reduced states coincide exactly. As explained there, this route stops at $k=3$: for $k>3$ the classification of finite multiply transitive permutation groups leaves only the alternating group A_n, which is too large to leave a nonzero invariant multiplicity subspace, and the Mathieu groups, which occur only at $n\in\{11,12,23,24\}$ and so give no family with $n\to\infty$ [65]. For fixed $k>3$ we therefore abandon exact symmetry and ask that a *typical* member of a suitable family of codes have k-site reduced states that are all close to one another, rather than exactly equal, and we show that such a typical code can be found and sampled efficiently.

b. A background-singlet family of codes. Let $\{e_1,\dots,e_d\}$ be the standard basis of $V_{\omega_1}=\mathbb{C}^d$, and let

$$|\zeta_d\rangle:=\frac{1}{\sqrt{d!}}\sum_{\pi\in S_d}\mathrm{sgn}(\pi)\,e_{\pi(1)}\otimes\cdots\otimes e_{\pi(d)}$$

be the normalized totally antisymmetric state of d qudits. This is the unique $SU(d)$-invariant vector in $(V_{\omega_1})^{\otimes d}$, and the reduced state of any $s\le d$ of its tensor factors is proportional to a projector onto an antisymmetric subspace (Lemma S3.35). In particular, every one-site reduced state is the maximally mixed state $\tau:=I_d/d$.

Throughout this section $n=qd+1$; see Section V B for the motivation of this choice. We partition the $n=qd+1$ physical positions into a distinguished position 0 and q blocks $B_1,\dots,B_q$ of size d, and define the seed isometry

$$\mathcal{V}_0\,|\psi\rangle:=|\psi\rangle_0\otimes|\zeta_d\rangle_{B_1}\otimes\cdots\otimes|\zeta_d\rangle_{B_q}.$$

This isometry places the logical qudit at site 0 and fills the rest of the system with q independent $SU(d)$-singlets, which we call *background sites*. It is exactly $SU(d)$-covariant (Proposition S3.36). Identifying $\mathcal{V}_0$ through the Schur–Weyl inclusion $W_{\omega_1}:V_{\omega_1}\otimes M_{\omega_1}\to(V_{\omega_1})^{\otimes n}$ gives a unit vector $|m_0\rangle\in M_{\omega_1}$ with $\mathcal{V}_0=W_{\omega_1}(I_{V_{\omega_1}}\otimes|m_0\rangle)$. For a general unit vector $|m\rangle\in M_{\omega_1}$, define

$$\mathcal{V}_m:=W_{\omega_1}(I_{V_{\omega_1}}\otimes|m\rangle),\qquad\mathcal{E}_m(X):=\mathcal{V}_mX\mathcal{V}_m^\dagger,$$

Every $\mathcal{V}_m$ is an exactly $SU(d)$-covariant isometric encoding, and $\mathcal{V}_0=\mathcal{V}_{m_0}$ is one particular member of this family, indexed by the multiplicity vector $|m\rangle$.

Averaging the encoding over a Haar-random $|m\rangle\in M_{\omega_1}$ reproduces the permutation twirl of the seed encoding:

$$\mathbb{E}_{m\sim\mathrm{Haar}(M_{\omega_1})}[\mathcal{V}_mX\mathcal{V}_m^\dagger]=\frac{1}{n!}\sum_{\pi\in S_n}P_\pi\,\mathcal{V}_0X\mathcal{V}_0^\dagger P_\pi^\dagger,$$

proved in Proposition S3.37. This identity is the source of the *full* permutation symmetry that an exactly k-transitive subgroup enforced directly in the construction for $k\le 3$.

c. The mean k-site channel. For a subset $S\subseteq\{1,\dots,n\}$ of size k, define the erased-subsystem channel

$$\Phi_{S,m}(X):=\mathrm{Tr}_{\hat{S}}(\mathcal{V}_mX\mathcal{V}_m^\dagger).$$

This is the reduced-state map $\rho^{(S)}$ of Eq. (9), written here as a linear map and for a general multiplicity vector m. Its Haar average over m defines the mean k-site channel $\overline{\Phi}_{k,n}:=\mathbb{E}_m\Phi_{S,m}$, which by the orbit-twirl identity above is, after the canonical identification of retained sites, independent of S. Proposition S3.40 evaluates it exactly:

$$\overline{\Phi}_{k,n}(X)=\frac{n-k}{n}\mathrm{Tr}(X)\,\beta_{k,N}+\frac{1}{n}\sum_{j=1}^{k}X^{(j)}\otimes\beta_{k-1,N}^{(\hat{j})},$$

where $N:=n-1=qd$ is the number of background sites and $\beta_{\ell,N}$ is the ℓ-site reduced state of the permutation twirl of the background, and Proposition S3.38 evaluates it as a sum over set partitions of the ℓ sample labels. Two sampled sites fall into the same singlet block with probability $O_{d,k}(1/n)$, so $\beta_{\ell,N}$ is close to $\tau^{\otimes\ell}$ in trace norm for fixed ℓ (Lemma S3.39), and $\overline{\Phi}_{k,n}(X)$ is therefore close to $\mathrm{Tr}(X)\,\tau^{\otimes k}$ up to a logical-dependent correction of order $1/n$, exactly as in the one-, two-, and three-site cases above.

Let

$$\lambda_{k,n}:=\overline{\Phi}_{k,n}(\tau),\quad\Lambda_{k,n}(X):=\mathrm{Tr}(X)\lambda_{k,n}.\tag{52}$$

Both $\overline{\Phi}_{k,n}$ and $\Lambda_{k,n}$ are $SU(d)$-covariant, since every $\beta_{\ell,N}$ is $SU(d)$-invariant, so by Lemma S2.2 the worst-case entanglement fidelity between them is attained on the maximally mixed logical input, and a direct expansion of the root fidelity around this point (Proposition S3.42) gives

$$F(\overline{\Phi}_{k,n},\Lambda_{k,n})=1-\frac{k(d^2-1)}{8n^2}+O_{d,k}(n^{-3}).$$

At $k=1$ and $k=3$ this reproduces, term by term, the coefficients found above by the exact symmetry-based calculation, Eq. (43) at $r=1$ and Eq. (51).

d. Haar-typical multiplicity vectors. The mean channel $\overline{\Phi}_{k,n}$ is an average over the whole family $\{\mathcal{E}_m\}$, not the channel produced by any single code. Turning this into a statement about one encoding requires a single unit vector $|m\rangle \in M_{\omega_1}$ for which *every* one of the $\binom{n}{k}$ erased-subsystem channels $\Phi_{S,m}$ is simultaneously close to $\overline{\Phi}_{k,n}$ in diamond norm $\|\cdot\|_\diamond$, the completely bounded trace norm on channels. Because the multiplicity space $M_{\omega_1} \cong [q+1, q^{d-1}]$ has dimension D_q given by Lemma S3.12, which grows exponentially in n for fixed d by the lower bound of Lemma S3.13, a Haar-random unit vector makes every matrix element of $\Phi_{S,m}$ concentrate sharply around its mean, and a union bound over the finitely many matrix elements and erased sets shows that a single random draw is simultaneously good for all of them, with probability approaching one exponentially fast in n.

Theorem 8. *Let $r_k := d^{k+1}$ and $M_{n,k} := \binom{n}{k} d^{2k+2}$. Let $|m\rangle$ be Haar-random in M_{ω_1}. Then*

$$\mathbb{P}\left[\max_{|S|=k} \|\Phi_{S,m} - \overline{\Phi}_{k,n}\|_\diamond > r_k^{3/2} D_q^{-1/4}\right] \le \frac{M_{n,k} D_q^{1/2}}{D_q + 1}.$$

For fixed d, k, the right-hand side is $e^{-\Omega_d(n)}$.

This is proved as Theorem S3.46. The argument writes each matrix element of $\Phi_{S,m} - \overline{\Phi}_{k,n}$ as a quadratic form in $|m\rangle$ (Lemma S3.43), bounds its variance using the Haar second-moment formula (Lemma S3.44), and converts the resulting entrywise bound into a diamond-norm bound through the unnormalized Choi matrix (Lemma S3.45), before taking a union bound over the $M_{n,k}$ matrix entries appearing across all $\binom{n}{k}$ erased sets.

A useful way to read this result is that the error-correcting property does not rely on fine-tuning a special multiplicity vector. Covariance first restricts the encoding to a fixed isotypic component, and the remaining freedom is the choice of a direction in its multiplicity space. Theorem 8 shows that, with respect to the natural Haar measure on this space, the set of multiplicity vectors for which some erased subsystem reveals an atypically large amount of logical information has exponentially small measure. Specially aligned and poorly performing vectors certainly exist: the seed vector $|m_0\rangle$ itself is one, since $\mathcal{V}_0$ leaves the logical qudit unencoded at site 0, so erasing that single site reveals the logical state completely. Such vectors are non-generic. Approximate decoupling from any fixed-size erased subsystem is therefore a typical property of the covariant sector rather than a feature of an isolated construction.

e. Multi-site erasure bound for a Haar-random code. Let $\mathcal{N}$ be the flagged k-site erasure channel (2) with a distribution $p = (p_S)_{|S|=k}$ on the k-element subsets of $\{1, \dots, n\}$. By Eq. (7), the complementary channel to $\mathcal{N} \circ \mathcal{E}_m$ is

$$\widehat{\mathcal{N} \circ \mathcal{E}_m}(\rho_L) = \sum_{|S|=k} p_S \, |S\rangle\langle S|_{F_E} \otimes \Phi_{S,m}(\rho_L).$$

As in the three-site case, the environment can access the logical state only through the reduced states on the erased set, here $\Phi_{S,m}(\rho_L)$ for every possible S. The difference is that these reduced states are no longer exactly equal to one another. Theorem 8 controls all of them simultaneously by the single mean channel $\overline{\Phi}_{k,n}$, and Proposition S3.42 controls how far that mean channel is from a constant one. Combining the two gives the erasure bound.

Theorem 9. *Let $d \ge 2$ and $k \ge 1$ be fixed. For every sufficiently large $n = qd+1$, let $|m\rangle$ be Haar-random in M_{ω_1} and let $\mathcal{E}_m$ be as above. With probability at least $1 - e^{-\Omega_d(n)}$ over $|m\rangle$, the following holds: for every distribution p on the k-element subsets of $\{1, \dots, n\}$ there exists a recovery channel $\mathcal{R}_p$ such that*

$$d\left(\mathcal{R}_p \circ \mathcal{N} \circ \mathcal{E}_m, \mathrm{id}\right) \le \frac{\sqrt{k(d^2-1)}}{2\sqrt{2}} \frac{1}{n} + O_{d,k}(n^{-2}).$$

In particular, with probability at least $1 - e^{-\Omega_d(n)}$ over the draw of $|m\rangle$, the code $\mathcal{E}_m$ is $O(\sqrt{k(d^2-1)}/n)$-correctable against flagged k-site erasure, uniformly over the erasure distribution p.

Proof. Condition on the event of Theorem 8, which has probability at least $1 - M_{n,k} D_q^{1/2}/(D_q+1) = 1 - e^{-\Omega_d(n)}$ and on which

$$\max_{|S|=k} \|\Phi_{S,m} - \overline{\Phi}_{k,n}\|_\diamond \le r_k^{3/2} D_q^{-1/4}.$$

We now introduce the two auxiliary channels

$$\overline{\Lambda}(\rho_L) := \sum_{|S|=k} p_S \, |S\rangle\langle S|_{F_E} \otimes \overline{\Phi}_{k,n}(\rho_L),$$

$$\Lambda_0(\rho_L) := \mathrm{Tr}(\rho_L) \sum_{|S|=k} p_S \, |S\rangle\langle S|_{F_E} \otimes \lambda_{k,n},$$

where $\lambda_{k,n}$ is taken from Eq. (52), so that Λ_0 is a constant channel in the sense of Eq. (4). The three channels $\widehat{\mathcal{N} \circ \mathcal{E}_m}$, $\overline{\Lambda}$, and Λ_0 are block diagonal with respect to the same flag decomposition and carry the same classical weights p_S. Root fidelity of two such block-diagonal states is the p-weighted sum of the block root fidelities, so for every reference-assisted input the Fuchs–van de Graaf inequality $1 - f(\rho, \sigma) \le \frac{1}{2}\|\rho - \sigma\|_1$ gives, block by block,

$$1 - F\left(\widehat{\mathcal{N} \circ \mathcal{E}_m}, \overline{\Lambda}\right) \le \frac{1}{2} \max_{|S|=k} \|\Phi_{S,m} - \overline{\Phi}_{k,n}\|_\diamond \le \frac{1}{2} r_k^{3/2} D_q^{-1/4},$$

while $d(\overline{\Lambda}, \Lambda_0) = d(\overline{\Phi}_{k,n}, \Lambda_{k,n})$. By Eq. (1), $d(\cdot,\cdot)$ is the supremum over purified logical inputs of the Bures-type state distance $\sqrt{1-f}$, and a supremum of metrics is a metric, so the triangle inequality gives

$$d\left(\widehat{\mathcal{N}\circ\mathcal{E}_m},\Lambda_0\right)\le\sqrt{\tfrac{1}{2}r_k^{3/2}D_q^{-1/4}}+d\left(\overline{\Phi}_{k,n},\Lambda_{k,n}\right).$$

The first term is $e^{-\Omega_d(n)}$, because D_q grows exponentially in n by Lemma S3.13 while r_k is fixed. By Eq. (S56) and Eq. (1), the second term equals $\sqrt{k(d^2-1)/8}\,n^{-1}+O_{d,k}(n^{-2})$. The complementary-channel theorem, Eq. (5), gives

$$\begin{aligned}\min_{\mathcal{R}_p} d\left(\mathcal{R}_p\circ\mathcal{N}\circ\mathcal{E}_m,\mathrm{id}\right)&=\min_{\Lambda\,\mathrm{constant}} d\left(\widehat{\mathcal{N}\circ\mathcal{E}_m},\Lambda\right)\\&\le d\left(\widehat{\mathcal{N}\circ\mathcal{E}_m},\Lambda_0\right),\end{aligned}$$

and the stated bound follows. □

f. Efficient sampling of multiplicity vectors. Theorem 8 shows that a typical multiplicity vector works, but sampling a Haar-random unit vector in M_{ω_1} naively requires about $D_q=e^{\Theta(n)}$ random bits, one for every amplitude in the Young–Yamanouchi basis of M_{ω_1} introduced in Section S3.3.2 (any other convenient basis will do). The proof of Theorem 8 only ever uses second and fourth moments of the quadratic forms $\langle m|\,A\,|m\rangle$ of Lemma S3.43, so full independence of the amplitudes is unnecessary and four-wise independence of their phases suffices to repeat the same concentration argument. This lets an exponentially long random string be replaced by a seed of length $O(n\log d)$, from which every phase can be computed efficiently.

Theorem 10. *Fix d and k, and let $n=qd+1$. There is an explicit randomized algorithm using $O(n\log d)$ random bits that outputs a seed s specifying a flat multiplicity vector*

$$|m_s\rangle=\frac{1}{\sqrt{D_q}}\sum_{T\in\mathrm{SYT}(q+1,q^{d-1})}z_s(T)\,|T\rangle,$$

where $z_s(T)\in\{1,-1,i,-i\}$ for all T, such that each phase $z_s(T)$ is computable in time $\mathrm{poly}(n,\log d)$, and

$$\mathbb{P}_s\left[\max_{|S|=k}\|\Phi_{S,m_s}-\overline{\Phi}_{k,n}\|_\diamond>r_k^{3/2}D_q^{-1/4}\right]\le\frac{M_{n,k}}{\sqrt{D_q}}=e^{-\Omega_d(n)}.$$

This is proved as Theorem S3.47. The construction labels every tableau T by its row word and assigns it a phase using a random cubic polynomial over $\mathbb{F}_{2^h}$, $h=n\log d$, evaluated at the tableau's label. The four coefficients of the polynomial form the seed s. A Vandermonde-determinant argument shows that the resulting phases are exactly four-wise independent, which is all the concentration bound of Theorem 8 requires. Only $4h=O(n\log d)$ random bits are used, and each phase is computable in $\mathrm{poly}(n,\log d)$ time.

This strengthens Theorem 8 in an operationally relevant way. Haar randomness is a useful benchmark because it shows that good multiplicity vectors are abundant, but writing down a generic Haar-random vector means specifying $D_q=e^{\Theta(n)}$ amplitudes, so by itself it does not give an efficient code-construction procedure. The bounded-independence construction replaces the full continuous randomness by a short classical seed from which every phase is generated efficiently. The resulting ensemble retains exactly the moment information the concentration argument uses, namely second and fourth moments of the quadratic forms in Lemma S3.43, while consuming only $O(n\log d)$ random bits and allowing each phase to be evaluated in $\mathrm{poly}(n,\log d)$ time. In this sense the construction identifies how much randomness the error-correcting property actually needs.

g. Multi-site erasure bound for an efficiently sampled code. Let $|m_s\rangle$ be the multiplicity vector produced by Theorem 10 and let $\mathcal{E}_s(X):=\mathcal{V}_{m_s}X\mathcal{V}_{m_s}^\dagger$. The proof of Theorem 9 uses the Haar distribution only through the diamond-norm bound of Theorem 8, which Theorem 10 reproduces for $|m_s\rangle$ with the same threshold $r_k^{3/2}D_q^{-1/4}$ and a failure probability that is again $e^{-\Omega_d(n)}$. The same argument therefore applies to this case with $|m\rangle$ replaced by $|m_s\rangle$.

Corollary 1. *Let $d\ge 2$ and $k\ge 1$ be fixed. For every sufficiently large $n=qd+1$, let $|m_s\rangle$ and $\mathcal{E}_s$ be as above. With probability at least $1-e^{-\Omega_d(n)}$ over the seed s, the following holds: for every distribution p on the k-element subsets of $\{1,\ldots,n\}$ there exists a recovery channel $\mathcal{R}_p$ such that*

$$d\left(\mathcal{R}_p\circ\mathcal{N}\circ\mathcal{E}_s,\mathrm{id}\right)\le\frac{\sqrt{k(d^2-1)}}{2\sqrt{2}}\frac{1}{n}+O_{d,k}(n^{-2}).$$

In particular, with probability at least $1-e^{-\Omega_d(n)}$ over the seed s, the code $\mathcal{E}_s$ is $O(\sqrt{k(d^2-1)}/n)$-correctable against flagged k-site erasure, uniformly over the erasure distribution p.

Proof. Condition on the event of Theorem 10 instead of that of Theorem 8. It has probability at least $1-M_{n,k}D_q^{-1/2}=1-e^{-\Omega_d(n)}$ and gives the same bound

$$\max_{|S|=k}\|\Phi_{S,m_s}-\overline{\Phi}_{k,n}\|_\diamond\le r_k^{3/2}D_q^{-1/4}.$$

The rest of the proof of Theorem 9 uses no other property of the multiplicity vector and applies unchanged. □

D. Arbitrary flagged multi-qudit noise

We now extend the erasure results to arbitrary flagged noise acting on a fixed number k of physical sites, the single-site case included. The argument does not depend on the value of k. The complementary channel (8) for arbitrary flagged noise on a set S depends on the logical input only through the reduced state $\rho^{(S)}(\rho_L)$ on that set. All that the argument uses is that this reduced state splits as

$$\rho^{(S)}(\rho_L)=\tau_S+\Delta_S(\rho_L),$$

with τ_S independent of the logical input and $\Delta_S = O_{d,k}(1/n)$ uniformly over the sets S on which the noise can act. This is Lemma S3.50, which turns such a uniform bound into an $O_{d,k}(1/\sqrt{n})$ correctability statement for arbitrary flagged noise supported on those sets.

What makes the reduced state alone sufficient is linearity. By Lemma S3.48, the complementary channel on the flag S is the composition $\widehat{\mathcal{N}}_S \circ \rho^{(S)}$, so the environment sees the logical state only after it has passed through the reduced-state map, whatever the local noise does. Both maps are linear, so the splitting survives the composition,

$$\widehat{\mathcal{N}}_S\left(\rho^{(S)}(\rho_L)\right) = \widehat{\mathcal{N}}_S(\tau_S) + \widehat{\mathcal{N}}_S\left(\Delta_S(\rho_L)\right),$$

and the logical-dependent term keeps the $1/n$ prefactor carried by Δ_S. Applying $\widehat{\mathcal{N}}_S$ cannot enlarge that term either, because the trace norm is contractive under channels. An unknown local noise channel therefore neither creates nor amplifies the dependence on the logical input, therefore it acts on a signal that is already suppressed by $1/n$, and the complementary channel approaches a constant one at the rate at which the reduced states do.

Two constructions of this paper supply the required uniform bound. For $k \leq 3$, the exactly transitive codes of Section V B make the reduced states on all k-tuples identical up to the canonical identification of sites, and their logical-state-dependent part is controlled by Eq. (48). For arbitrary fixed k, the codes of Section V C give the same bound with the exact equality replaced by the diamond-norm estimate of Theorem 8.

This allows one to compare the complementary channel of arbitrary channel with a constant channel obtained by replacing each reduced state by its input-independent part. The proof uses a trace-norm estimate together with the Fuchs–van de Graaf inequality, rather than an explicit fidelity calculation. The resulting bounds are stated in Theorems S3.51 and S3.53:

Theorem 11. *Let $\mathcal{H}_L = V_{\omega_1}$ and $\mathcal{H}_P = (V_{\omega_1})^{\otimes n}$. Assume that $d = p^r$, where p is prime and r is a positive integer, and that $n-1 = p^s$, where $s \geq r$ is a positive integer. Equip $\mathcal{H}_L$ with the fundamental representation of $\mathfrak{su}(d)$ and $\mathcal{H}_P$ with the transversal representation*

$$t_a \longmapsto \sum_{i=1}^{n} t_a^{(i)}, \qquad a = 1, \ldots, d^2-1,$$

where $t_a^{(i)}$ denotes the fundamental action of t_a on the ith physical space V_{ω_1}. For sufficiently large n, there exists an $\mathfrak{su}(d)$-covariant encoding $\mathcal{E}$ with respect to these representations whose code space is invariant under the action of $\mathrm{PGL}(2, n-1)$. For any flagged noise channel supported on sets of at most three sites,

$$\mathcal{N}(\sigma) = \sum_{1 \leq |S| \leq 3} p_S \left|S\right\rangle \left\langle S\right|_F \otimes \left(\mathcal{N}_S \otimes \mathrm{id}_{\hat{S}}\right)(\sigma),$$

with arbitrary channels $\mathcal{N}_S$ acting on the sites of S, there exists a recovery channel $\mathcal{R}$ such that

$$d(\mathcal{R}\mathcal{N}\mathcal{E}, \mathrm{id}) = O_d\left(\frac{1}{\sqrt{n}}\right).$$

In particular, the code is $O_d(1/\sqrt{n})$-correctable against arbitrary flagged noise on any set of at most three qudits, single-qudit noise included, uniformly over the distribution of noisy sets.

Theorem 12. *Let $d \geq 2$ and $k \geq 1$ be fixed, let $n = qd+1$, and let $\mathcal{E}_s$ be the efficiently sampled $SU(d)$-covariant encoding of Theorem 10. With probability at least $1 - e^{-\Omega_d(n)}$ over the seed s, the following holds: for every flagged k-qudit noise channel*

$$\mathcal{N}(\sigma) = \sum_{|S|=k} p_S \left|S\right\rangle \left\langle S\right|_F \otimes \left(\mathcal{N}_S \otimes \mathrm{id}_{\hat{S}}\right)(\sigma),$$

with arbitrary channels $\mathcal{N}_S$ acting on the erased sites, there exists a recovery channel $\mathcal{R}$ such that

$$d(\mathcal{R}\mathcal{N}\mathcal{E}_s, \mathrm{id}) \leq \sqrt{\frac{k}{n}} + e^{-\Omega_d(n)}.$$

In particular, with probability at least $1 - e^{-\Omega_d(n)}$ over the seed s, the code $\mathcal{E}_s$ is $O(\sqrt{k/n})$-correctable against arbitrary flagged k-site noise, uniformly over the distribution of noisy sets.

The loss from the $1/n$ erasure scaling to the $1/\sqrt{n}$ scaling comes from the use of the trace-norm bound and the Fuchs–van de Graaf inequality in place of an explicit fidelity calculation. The argument applies to arbitrary local noise channels, but it is not expected to give sharp constants or optimal scaling in general. A direct fidelity calculation for more general noise models is left for future work.

VI. COVARIANT ENCODING FOR ANALOG QUANTUM SIMULATION

We now discuss how the covariant AQECCs constructed in Sections IV and V can be used for analog quantum simulation. The aim is to run the simulated evolution directly on the encoded state: each generator of the target dynamics has to be implemented on the code space by an operation the physical system already performs, here a sum of single-site generators, so that the encoding intertwines the simulated dynamics with the physical one. This is the covariance condition of Section III B, and the Eastin–Knill theorem is what forces it to hold only approximately. Applied to every Hamiltonian on the simulated Hilbert space $\mathcal{H}$, it would require covariance with respect to all of $\mathfrak{su}(\mathcal{H})$.

In analog simulation, however, one usually does not need to implement all unitaries on the full Hilbert space.

Instead, the available evolutions are generated by a restricted set of Hamiltonians, and the Lie algebra generated by these Hamiltonians is the dynamical Lie algebra (DLA). In many symmetric physical systems, the DLA is contained in an algebra of the form

$$\mathfrak{su}(d_1) \oplus \cdots \oplus \mathfrak{su}(d_k) \oplus \mathfrak{u}(1)^{\oplus(r-1)}.$$

An AQECC covariant with respect to this smaller algebra therefore makes all dynamics generated within it act transversally on the encoded space.

In this section, we construct such codes by building on the $SU(d)$-covariant AQECCs from the previous sections. We call the resulting construction a block encoding. We then analyze its performance under two-qudit erasure noise, including both in-block and inter-block erasures. For a schematic illustration see Fig. 1(d).

A. Dynamical Lie algebra and Symmetries

a. Symmetric analog dynamics. Consider a quantum system consisting of N qubits, with Hilbert space

$$\mathcal{H} = (\mathbb{C}^2)^{\otimes N}, \qquad \dim \mathcal{H} = d = 2^N.$$

The dynamics are governed by a time-dependent Hamiltonian

$$H(t) = H_0 + \sum_{j=1}^{m} u_j(t) H_j, \tag{53}$$

where H_0 is the intrinsic drift Hamiltonian, $H_j{}_{j=1}^{m}$ are control Hamiltonians, and $u_j(t) \in \mathbb{R}$ are tunable control fields.

In this section, we focus on *global controls.* By this we mean controls that act collectively on the physical sites and preserve the relevant permutation symmetry. Collective spin operators give a standard example:

$$\sum_{i=1}^{N} \sigma_\alpha^{(i)}, \qquad \alpha = x, y, z.$$

Such controls are invariant under permutations of the physical sites. In Section VII, we extend the discussion to *local controls*, which act on individual sites and generally break the permutation symmetry.

The set of reachable unitary transformations is determined by the *dynamical Lie algebra* [14, 67]

$$\mathrm{Lie}_{\mathbb{R}}\{iH_0, iH_1, \ldots, iH_m\} \subseteq \mathfrak{su}(\mathcal{H}).$$

Here $\mathfrak{su}(\mathcal{H})$ denotes the Lie algebra of traceless skew-Hermitian linear operators on $\mathcal{H}$. After choosing a basis of $\mathcal{H}$, this is the usual matrix Lie algebra $\mathfrak{su}(d)$.

The DLA depends on the specific Hamiltonians in Eq. (53). In general, determining it can be difficult. In many symmetric systems, however, the DLA is contained in an invariant Lie algebra associated with a finite site-permutation symmetry. This invariant algebra has a canonical block structure, which is the starting point for the block-encoding construction below.

b. Invariant Lie algebra. Let $G \subseteq S_N$ be a finite group acting on $\mathcal{H}$ by permuting the qubit sites:

$$P_g |i_1\rangle \otimes \cdots \otimes |i_N\rangle = \left|i_{g^{-1}(1)}\right\rangle \otimes \cdots \otimes \left|i_{g^{-1}(N)}\right\rangle.$$

Here G acts on the logical simulation Hilbert space. This should be distinguished from the finite permutation groups used earlier in Eq. (22), which acted on the physical subsystems of a code in order to impose symmetry constraints on the code space.

The induced action on operators is by conjugation, as in Eq. (25). The G-invariant linear operators are

$$\mathrm{End}(\mathcal{H})^G := X \in \mathrm{End}(\mathcal{H}) : P_g X P_g^\dagger = X \text{ for all } g \in G.$$

The corresponding *invariant Lie algebra* [17, 68] is

$$\mathfrak{su}(\mathcal{H})^G := \mathrm{End}(\mathcal{H})^G \cap \mathfrak{su}(\mathcal{H}) \subseteq \mathfrak{su}(\mathcal{H}).$$

In physical examples, G is usually a site-permutation symmetry of the Hamiltonians. For instance, if the drift and control Hamiltonians have the form

$$H_x = \sum_{i=1}^{N} \sigma_x^{(i)}, \qquad H_z = \sum_{i=1}^{N} \sigma_z^{(i)},$$
$$H_{zz} = \sum_{(i,j)\in E(\Gamma)} \sigma_z^{(i)} \sigma_z^{(j)},$$

where Γ is an interaction graph, then the site-permutation symmetry is the automorphism group of Γ. For a complete graph this group is S_N, while for a cycle graph it contains the dihedral symmetry D_N. Table VI A 0 b lists several standard examples of spin systems with finite site-permutation symmetries.

If all Hamiltonians in Eq. (53) are invariant under the action of G, then

$$iH_0, iH_1, \ldots, iH_m \in \mathfrak{su}(\mathcal{H})^G.$$

Since $\mathfrak{su}(\mathcal{H})^G$ is closed under commutators, the DLA is contained in the invariant Lie algebra:

$$\mathrm{Lie}_{\mathbb{R}}\{iH_0, iH_1, \ldots, iH_m\} \subseteq \mathfrak{su}(\mathcal{H})^G.$$

Thus $\mathfrak{su}(\mathcal{H})^G$ provides a symmetry-determined ambient algebra for the reachable dynamics. We do not attempt to characterize when the DLA is equal to $\mathfrak{su}(\mathcal{H})^G$. This controllability question depends on the detailed structure of the Hamiltonians and is known only in special cases; see Refs. [17, 69, 70].

c. Block structure from representation theory. As shown in Lemma S4.1, the invariant Lie algebra has the reductive decomposition

$$\mathfrak{su}(\mathcal{H})^G \cong \mathfrak{su}(d_1) \oplus \cdots \oplus \mathfrak{su}(d_k) \oplus \mathfrak{u}(1)^{\oplus(r-1)}. \tag{54}$$

This decomposition follows from the decomposition of $\mathcal{H}$ into irreducible representations of G:

$$\mathcal{H} \cong \bigoplus_{\alpha \in \mathcal{I}_{\mathcal{H}}} V_\alpha \otimes M_\alpha. \tag{55}$$

Physical system	Controlled Hamiltonian $H(t) = H_0 + \sum_{j=1}^{k} u_j(t) H_j$	Site-permutation symmetry	References
Uniform open spin chain	$\sum_{i=1}^{N-1} \sigma_z^{(i)} \sigma_z^{(i+1)} + u_x(t) S_x + u_z(t) S_z$	Reflection symmetry $C_2 \cong \mathbb{Z}_2$	[18, 19, 69]
Uniform spin ring	$\sum_{i=1}^{N} \sigma_z^{(i)} \sigma_z^{(i+1)} + u_x(t) S_x + u_z(t) S_z, \quad \sigma_z^{(N+1)} = \sigma_z^{(1)}$	Dihedral group D_N; translations give C_N	[19, 69, 70]
Spin network on a graph Γ	$\sum_{(i,j)\in E(\Gamma)} J_{ij}\, \sigma_z^{(i)} \sigma_z^{(j)} + u_x(t) S_x + u_z(t) S_z$	Weighted graph automorphism group $\mathrm{Aut}(\Gamma, J)$	[15, 19, 69, 70]
Fully connected collective-spin model	$S_z^2 + u_x(t) S_x + u_z(t) S_z$	Full permutation group S_N	[15, 69]
Tavis–Cummings model	$\omega(t) a^\dagger a + \frac{\hbar \Omega(t)}{2} S_z + \frac{g(t) \hbar \sqrt{N}}{2} (a + a^\dagger) S_x$	Permutation symmetry S_N of identical atoms	[13]
Rydberg atom array with uniform drive	$\frac{\Omega(t)}{2} S_x - \Delta(t) \sum_i n_i + \sum_{i<j} V_{ij} n_i n_j$	Geometric automorphisms of the array preserving V_{ij}	[6, 7]

Table I. Examples of spin Hamiltonian systems with finite site-permutation symmetries. In each row, both the drift Hamiltonian and the listed global controls are invariant under the corresponding group $G \subseteq S_N$. Hence, the DLA generated by these Hamiltonians is contained in $\mathfrak{su}(\mathcal{H})^G$. Here $S_\alpha = \sum_i \sigma_\alpha^{(i)}$, $\alpha = x, y, z$, and $n_i = \left(\sigma_z^{(i)} + 1\right)/2$.

Here V_α is an irreducible representation of G, M_α is the corresponding multiplicity space, and $\mathcal{I}_\mathcal{H}$ is the set of irreducible representations of G that appear in $\mathcal{H}$.

By definition, $\mathfrak{su}(\mathcal{H})^G$ consists of traceless skew-Hermitian operators on $\mathcal{H}$ that commute with the action of G. Schur's lemma implies that every such operator acts as the identity on each irrep factor V_α and acts non-trivially only on the corresponding multiplicity space M_α. Equivalently,

$$\mathrm{End}(\mathcal{H})^G \cong \bigoplus_{\alpha \in \mathcal{I}_\mathcal{H}} I_{V_\alpha} \otimes \mathrm{End}(M_\alpha).$$

After imposing skew-Hermiticity and the global tracelessness condition, this gives

$$\mathfrak{su}(\mathcal{H})^G \cong \bigoplus_{\alpha \in \mathcal{I}_\mathcal{H}} \mathfrak{su}(M_\alpha) \oplus \mathfrak{u}(1)^{\oplus(|\mathcal{I}_\mathcal{H}|-1)}.$$

Thus, if $m_\alpha := \dim M_\alpha$, then

$$\mathfrak{su}(\mathcal{H})^G \cong \mathfrak{su}(m_1) \oplus \cdots \oplus \mathfrak{su}(m_{|\mathcal{I}_\mathcal{H}|}) \oplus \mathfrak{u}(1)^{\oplus(|\mathcal{I}_\mathcal{H}|-1)}.$$

This identifies the parameters in Eq. (54): the numbers d_i are the dimensions of the multiplicity spaces associated with the irreducible representations of G appearing in $\mathcal{H}$, and $r = |\mathcal{I}_\mathcal{H}|$. In what follows, we will also write $d_i = m_i$.

B. $(\oplus_i \mathfrak{su}(d_i)) \oplus (\oplus_j \mathfrak{u}(1))$-covariant block encoding

A fully $\mathfrak{su}(\mathcal{H})$-covariant encoding would make every Hamiltonian on $\mathcal{H}$ act transversally on the code space. For analog simulation with symmetries, this is usually more covariance than is needed. If $\dim \mathcal{H} = d$, then a nontrivial transversal implementation of $SU(d)$ requires each physical subsystem to carry a nontrivial representation of $SU(d)$. Since the smallest nontrivial irreducible representation of $SU(d)$ is the fundamental representation, the local physical dimension is at least d. Thus, a fully $\mathfrak{su}(\mathcal{H})$-covariant encoding inherits the full Hilbert-space dimension of the simulated system.

In the symmetry-adapted setting of Section VI A, the reachable dynamics are contained in the smaller algebra

$$\mathfrak{su}(d_1) \oplus \cdots \oplus \mathfrak{su}(d_k) \oplus \mathfrak{u}(1)^{\oplus(r-1)}.$$

It is therefore sufficient to construct a code covariant with respect to this algebra, rather than with respect to all of $\mathfrak{su}(\mathcal{H})$. We do this by encoding each non-abelian block separately, using the $SU(d)$-covariant codes constructed above. This gives a block encoding adapted to the symmetry decomposition of the invariant algebra containing the DLA.

a. Definition of the block encoding. For the ath non-abelian block, let V_{d_a,λ_a} be an irreducible representation of $\mathfrak{su}(d_a)$, chosen as in Section V, and define

$$A_a := (V_{d_a,\lambda_a})^{\otimes n_a}, \qquad a = 1, \ldots, k.$$

For the abelian factors, choose physical spaces

$$B_s := (W_s)^{\otimes n_{k+s}}, \qquad s = 1, \ldots, r-1.$$

The total physical Hilbert space of the block encoding is

$$\mathcal{H}_P^{\text{block}} := A_1 \otimes \cdots \otimes A_k \otimes B_1 \otimes \cdots \otimes B_{r-1}.$$

The physical representation of

$$\mathfrak{su}(d_1) \oplus \cdots \oplus \mathfrak{su}(d_k) \oplus \mathfrak{u}(1)^{\oplus(r-1)}$$

is defined blockwise. For

$$(t_1, \ldots, t_k, u_1, \ldots, u_{r-1}) \in \mathfrak{su}(d_1)\oplus\cdots\oplus\mathfrak{su}(d_k)\oplus\mathfrak{u}(1)^{\oplus(r-1)},$$

we set

$$\begin{aligned}(t_1, \ldots, t_k, u_1, \ldots, u_{r-1}) &\longmapsto \\ &\pi_{d_1,\lambda_1}(t_1) + \cdots + \pi_{d_k,\lambda_k}(t_k) \\ &+ R_1(u_1) + \cdots + R_{r-1}(u_{r-1}).\end{aligned} \tag{56}$$

Here each term acts nontrivially only on its corresponding block and as the identity on all other blocks. The non-abelian block actions are transversal:

$$\pi_{d_j,\lambda_j}(t_j) = \sum_{i=1}^{n_j} \pi_{d_j,\lambda_j}^{(i+\sum_{s=1}^{j-1} n_s)}(t_j). \tag{57}$$

The operator $R_s(u_s)$ denotes the chosen representation of the generator $u_s \in \mathfrak{u}(1)$ on the abelian block $B_s = (W_s)^{\otimes n_{k+s}}$.

We do not analyze $U(1)$-covariant codes in this work. The abelian blocks should therefore be viewed as placeholders for any suitable $U(1)$-covariant construction. Such constructions can be obtained, for example, from Ref. [47], and can be used to specify the choices of W_s and R_s.

The resulting code space is a product code:

$$C_1 \otimes \cdots \otimes C_{k+r-1}. \tag{58}$$

For $i \leq k$, the factor C_i is the code space of an $\mathfrak{su}(d_i)$-covariant encoding. For $k < i \leq k+r-1$, the factor C_i is the code space of a $\mathfrak{u}(1)$-covariant encoding. This product structure is natural from representation theory, because irreducible representations of a direct sum of Lie algebras are tensor products of irreducible representations of the summands.

As a simple example, consider an $\mathfrak{su}(2) \oplus \mathfrak{su}(3)$-covariant encoding with

$$\mathcal{H}_P^{\text{block}} = (V_j)^{\otimes n} \otimes (V_{3,\omega_1})^{\otimes n},$$

where V_j is the spin-j representation of $\mathfrak{su}(2)$ and V_{3,ω_1} is the fundamental representation of $\mathfrak{su}(3)$. If λ_1 denotes the first standard Gell-Mann matrix, then the generator (σ_x, λ_1) is represented as

$$\sum_{i=1}^{n} J_x^{(i)} + \sum_{i=1}^{n} \lambda_1^{(i+n)}.$$

b. Encoding symmetry-adapted Hamiltonians. We now relate this block encoding to analog simulation. The Hamiltonians $H_0, H_1, \ldots, H_m$ need not be block diagonal in the computational basis. However, if G is a site-permutation symmetry of the Hamiltonians, then they become block diagonal in a symmetry-adapted basis associated with the decomposition (55). A systematic construction of such a basis using Young symmetrizers was given in Ref. [68].

Let U_{sym} denote the corresponding symmetry-adapted transform. Then every G-invariant Hamiltonian has the block form

$$iH \longmapsto U_{\text{sym}}^\dagger iHU_{\text{sym}} = \underbrace{i\widetilde{H}_\alpha}_{\in\mathfrak{su}(d_1)} \oplus \underbrace{i\widetilde{H}_\beta}_{\in\mathfrak{su}(d_2)} \oplus \cdots.$$

Once this decomposition is known, each block Hamiltonian is encoded using the corresponding block representation in Eq. (56). In Example S4.6, we work this out for $G = S_3$. In that case the relevant invariant Lie algebra is

$$\mathfrak{su}(4) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1) \subseteq \mathfrak{su}(2^3),$$

and we give the explicit block diagonal form of

$$\begin{gathered} H_x = \sigma_x^{(1)} + \sigma_x^{(2)} + \sigma_x^{(3)}, \qquad H_y = \sigma_y^{(1)} + \sigma_y^{(2)} + \sigma_y^{(3)}, \\ H_{zz} = \sigma_z^{(1)}\sigma_z^{(2)} + \sigma_z^{(1)}\sigma_z^{(3)} + \sigma_z^{(2)}\sigma_z^{(3)}. \end{gathered} \tag{59}$$

c. Resource scaling. The possible advantage of the block encoding is clearest for large symmetry groups. For $G = S_N$, Lemma S4.4 gives

$$\dim\left(\mathfrak{su}(\mathcal{H})^{S_N}\right) = \binom{N+3}{3} - 1.$$

A related result was obtained in Ref. [16]; we include the derivation in the Appendix for completeness. Therefore, the invariant Lie algebra has dimension polynomial in N, whereas

$$\dim \mathfrak{su}(\mathcal{H}) = 4^N - 1$$

for $\mathcal{H} = (\mathbb{C}^2)^{\otimes N}$.

For a general subgroup $G \subseteq S_N$, Lemma S4.5 shows that

$$\mathfrak{su}(\mathcal{H})^{S_N} \subseteq \mathfrak{su}(\mathcal{H})^G.$$

Consequently,

$$\dim \mathfrak{su}(\mathcal{H})^G \geq \dim \mathfrak{su}(\mathcal{H})^{S_N} = \binom{N+3}{3} - 1. \tag{60}$$

Larger symmetry groups impose more constraints and can therefore lead to smaller invariant Lie algebras. Compared with a fully $\mathfrak{su}(\mathcal{H})$-covariant encoding, the symmetry-adapted block encoding reduces the dimensions of the non-abelian blocks that need to be encoded covariantly. The amount of reduction depends on the symmetry group and on the multiplicities in the decomposition (55).

There are two main reasons to consider block encodings covariant with respect to

$$\mathfrak{su}(d_1) \oplus \cdots \oplus \mathfrak{su}(d_k) \oplus \mathfrak{u}(1)^{\oplus(r-1)}$$

rather than fully $\mathfrak{su}(\mathcal{H})$-covariant encodings. First, in the presence of symmetries, passing from the full algebra $\mathfrak{su}(\mathcal{H})$ to the invariant algebra $\mathfrak{su}(\mathcal{H})^G$ can reduce the size of the non-abelian sectors that must be protected covariantly. Second, from a practical perspective, stacking several $\mathfrak{su}(d_i)$-covariant code blocks may scale more favorably than increasing the local physical dimension to match the full dimension of $\mathcal{H}$, as would be required for a fully $\mathfrak{su}(\mathcal{H})$-covariant encoding.

In the block-encoding setting, the numbers n_i of physical qudits assigned to the different $\mathfrak{su}(d_i)$ blocks can be chosen independently. This gives additional flexibility in distributing physical resources among the different symmetry sectors. By contrast, in the fully $\mathfrak{su}(\mathcal{H})$-covariant setting, one is tied to a single global code block with a fixed global block size n. Therefore, the block encoding can reduce the required local physical dimension while retaining transversal implementation of the symmetry-preserving dynamics.

These advantages are useful only if the resulting block encoding still has good approximate error-correction properties. We analyze this question next.

C. Correlated flagged two-qudit erasure

a. From product noise to correlated erasures. For product noise across the blocks, the block code inherits the AQEC properties of its individual components. Indeed, consider a noise channel of the form

$$\mathcal{N} = \bigotimes_{i=1}^{2|\mathcal{I}_{\mathcal{H}}|-1} \mathcal{N}_i$$

for the

$$\mathfrak{su}(m_1) \oplus \cdots \oplus \mathfrak{su}(m_{|\mathcal{I}_{\mathcal{H}}|}) \oplus \mathfrak{u}(1)^{\oplus(|\mathcal{I}_{\mathcal{H}}|-1)}$$

-covariant block code constructed in Section VI B. Since the block code is a product code (58), and each $\mathcal{N}_i$ acts on a single block, the correctability analysis reduces to the corresponding analysis for the individual covariant codes. In particular, when the $\mathcal{N}_i$ are erasure-type channels of the form considered in Eqs. (2) and (3), the block code is corrected blockwise.

This product noise model does not capture correlated erasures involving qudits from different blocks. We therefore consider a two-qudit erasure model that includes both in-block and inter-block erasures.

b. Two-block code and reduced states. We focus on an $\mathfrak{su}(d_1) \oplus \mathfrak{su}(d_2)$-covariant block encoding. The two blocks have sizes n_1 and n_2, and the physical Hilbert space is

$$\mathcal{H}_P \cong (\mathbb{C}^{d_1})^{\otimes n_1} \otimes (\mathbb{C}^{d_2})^{\otimes n_2}.$$

Each physical qudit carries the fundamental representation of the corresponding Lie algebra. Therefore, the transversal physical action is

$$t_a \longmapsto \sum_{i=1}^{n_1} t_a^{(i)}, \qquad s_b \longmapsto \sum_{j=1}^{n_2} s_b^{(j)},$$

where $t_a^{(i)}$ acts on the ith qudit of the first block and $s_b^{(j)}$ acts on the jth qudit of the second block. For each block, we choose the code space as in Theorem 5, so that two-qudit reduced states inside each block are identical up to the canonical identification of sites. The precise number-theoretic conditions on n_1 and n_2 are those of Theorem 5; below we only use that both n_1 and n_2 can be taken arbitrarily large.

It is convenient to write a logical state in a form adapted to the two-block structure:

$$\rho_L = \frac{I}{d_1 d_2} + \frac{1}{d_2}\sum_a r_{1,a}, \bar{t}_a \otimes I + \frac{1}{d_1}\sum_b r_{2,b}, I \otimes \bar{s}_b + \sum_{a,b} C_{ab}, \bar{t}_a \otimes \bar{s}_b.$$

Here $r_{1,a}$ and $r_{2,b}$ are the Bloch-vector components of the one-block reduced logical states, and C_{ab} is the inter-block correlation matrix. The barred generators act on the logical spaces.

The relevant two-qudit reduced states fall into three classes: two erased qudits in the first block, two erased qudits in the second block, and one erased qudit in each block. The detailed derivation is given in Proposition S4.7. For two sites i, j in the first block, the reduced state is

$$\rho_{1,\mathrm{in}}^{(ij)}(\rho_L) = \frac{1}{d_1^2}\mathbf{1} - \frac{2}{d_1 n_1}\sum_{m=1}^{d_1^2-1} t_m^{(i)} t_m^{(j)} + \frac{1}{n_1 d_1}\sum_{b=1}^{d_1^2-1} r_{1,b}\left(t_b^{(i)} + t_b^{(j)}\right).$$

For two sites i, j in the second block,

$$\rho_{2,\mathrm{in}}^{(ij)}(\rho_L) = \frac{1}{d_2^2}\mathbf{1} - \frac{2}{d_2 n_2}\sum_{m=1}^{d_2^2-1} s_m^{(i)} s_m^{(j)} + \frac{1}{n_2 d_2}\sum_{b=1}^{d_2^2-1} r_{2,b}\left(s_b^{(i)} + s_b^{(j)}\right).$$

These in-block expressions can be obtained by tracing out one site from the three-qudit reduced state in Eq. (48).

For an inter-block pair, with i in the first block and j in the second block, the reduced state is

$$\rho_{\text{inter}}^{(ij)}(\rho_L) = \frac{I}{d_1 d_2} + \frac{1}{n_1 d_2}\sum_a r_{1,a}, t_a^{(i)} \otimes I + \frac{1}{n_2 d_1}\sum_b r_{2,b}, I \otimes s_b^{(j)} + \frac{1}{n_1 n_2}\sum_{a,b} C_{ab}, t_a^{(i)} \otimes s_b^{(j)}.$$

This follows from the product form of the block encoding and the one-site reduced-state formula (42).

c. Flagged two-qudit erasure. erasure channel. Let $n = n_1 + n_2$, where the first block consists of sites $1, \dots, n_1$ and the second block consists of sites $n_1 + 1, \dots, n$. The flagged two-qudit erasure channel is

$$\mathcal{N}(\sigma) = \sum_{1 \le i < j \le n} q_{ij} \left|ij\right\rangle \left\langle ij\right|_F \otimes \left|e\right\rangle \left\langle e\right|_{ij} \otimes \operatorname{Tr}_{ij}(\sigma),$$

where $q_{ij} \ge 0$, $\sum_{i<j} q_{ij} = 1$, and the flag register records the erased pair.

There are three types of erased pairs: two sites in the first block, two sites in the second block, and one site in each block. We denote the corresponding total probabilities by

$$p_1 := \sum_{1 \le i < j \le n_1} q_{ij}, \qquad p_2 := \sum_{n_1 < i < j \le n} q_{ij}, \qquad p_{12} := \sum_{1 \le i \le n_1 \; n_1 < j \le n} q_{ij}.$$

Thus $p_{12} = 1 - p_1 - p_2$. The remaining information about which pair was erased is stored in orthogonal flag states, denoted by $\omega_{1,\text{in}}$, $\omega_{2,\text{in}}$, and ω_{inter}, supported on the three corresponding flag subspaces.

Using the permutation symmetries imposed separately on the two blocks, the reduced states within each of the three sectors can be identified up to relabeling of sites. Therefore, the complementary channel has the form

$$\widehat{\mathcal{N} \circ \mathcal{E}}(\rho_L) = p_1, \omega_{1,\text{in}} \otimes \rho_{1,\text{in}}(\rho_L) + p_2, \omega_{2,\text{in}} \otimes \rho_{2,\text{in}}(\rho_L) + p_{12}, \omega_{\text{inter}} \otimes \rho_{\text{inter}}(\rho_L).$$

Here the site labels have been suppressed because the reduced states are identified within each sector.

We compare this complementary channel with the constant channel

$$\Lambda_0(\rho) = \operatorname{Tr}(\rho) \left(p_1 \omega_{1,\text{in}} \otimes \tau_1 + p_2 \omega_{2,\text{in}} \otimes \tau_2 + p_{12} \omega_{\text{inter}} \otimes \tau_{12}\right),$$

where

$$\tau_1 := \frac{I}{d_1^2} - \frac{2}{d_1 n_1}\sum_m t_m^{(i)} t_m^{(j)}, \qquad \tau_2 := \frac{I}{d_2^2} - \frac{2}{d_2 n_2}\sum_m s_m^{(i)} s_m^{(j)}, \qquad \tau_{12} := \frac{I}{d_1 d_2}.$$

These are the logical-input-independent parts of the first in-block, second in-block, and inter-block reduced states, respectively.

The explicit fidelity calculation in Proposition S4.7 gives

$$F\left(\widehat{\mathcal{N} \circ \mathcal{E}}, \Lambda_0\right) = 1 - \frac{p_1(d_1^2 - 1)}{4n_1^2} - \frac{p_2(d_2^2 - 1)}{4n_2^2} - p_{12}\left(\frac{d_1^2 - 1}{8n_1^2} + \frac{d_2^2 - 1}{8n_2^2}\right) + O_{d_1,d_2}\left(n_1^{-2} n_2^{-2} + n_1^{-3} + n_2^{-3}\right). \tag{61}$$

Together with the complementary-channel characterization of AQEC, this gives the following result.

Theorem 13. *Let*

$$\mathcal{H}_L = V_{d_1,\omega_1} \otimes V_{d_2,\omega_1}, \qquad \mathcal{H}_P = (V_{d_1,\omega_1})^{\otimes n_1} \otimes (V_{d_2,\omega_1})^{\otimes n_2}.$$

Assume that each n_ℓ is a prime power satisfying $n_\ell \equiv 1 \pmod{d_\ell}$, for $\ell = 1, 2$. Equip $\mathcal{H}_L$ with the product fundamental representation of $\mathfrak{su}(d_1) \oplus \mathfrak{su}(d_2)$ and $\mathcal{H}_P$ with the transversal representation

$$t_a \longmapsto \sum_{i=1}^{n_1} t_a^{(i)}, \qquad s_b \longmapsto \sum_{j=1}^{n_2} s_b^{(j)}.$$

Let $\mathcal{E}$ be an $\mathfrak{su}(d_1) \oplus \mathfrak{su}(d_2)$-covariant encoding with respect to these representations, such that each block is invariant under the action of $\mathrm{AGL}(1, n_\ell)$ (46)*. Then, for the flagged two-qudit erasure channel*

$$\mathcal{N}(\sigma) = \sum_{1 \le i < j \le n_1 + n_2} q_{ij} \left|ij\right\rangle \left\langle ij\right|_F \otimes \left|e\right\rangle \left\langle e\right|_{ij} \otimes \operatorname{Tr}_{ij}(\sigma),$$

there exists a recovery channel $\mathcal{R}$ such that, when $n_1 \approx n_2 \approx n$,

$$d(\mathcal{R}\mathcal{N}\mathcal{E}, \mathrm{id}) \le \frac{\sqrt{C_1(d_1^2 - 1) + C_2(d_2^2 - 1)}}{n} + O_{d_1,d_2}(n^{-2}),$$

where C_1 and C_2 are constants independent of the encoding parameters. In particular, the code is

$$O\left(\frac{\sqrt{C_1(d_1^2 - 1) + C_2(d_2^2 - 1)}}{n}\right)$$

-correctable against flagged erasure on two sites.

Proof. By Eq. (61) and the definition of the channel distance in Eq. (1), the distance between the complementary channel and Λ_0 is bounded by

$$d\left(\widehat{\mathcal{N} \circ \mathcal{E}}, \Lambda_0\right) \le \frac{\sqrt{C_1(d_1^2 - 1) + C_2(d_2^2 - 1)}}{n} + O_{d_1,d_2}(n^{-2})$$

for $n_1 \approx n_2 \approx n$, with constants C_1 and C_2 independent of the encoding parameters. The complementary-channel theorem, Eq. (5), gives

$$\min_{\mathcal{R}} d(\mathcal{R}\mathcal{N}\mathcal{E}, \mathrm{id}) = \min_{\Lambda,\text{constant}} d\left(\widehat{\mathcal{N} \circ \mathcal{E}}, \Lambda\right) \le d\left(\widehat{\mathcal{N} \circ \mathcal{E}}, \Lambda_0\right),$$

and the stated bound follows. $\square$

As a special case, the same result gives the two-qudit erasure analysis for the usual in-block $SU(d)$-covariant encoding, as discussed in Section V B.

VII. COVARIANT ENCODING FOR UNIVERSAL ANALOG COMPUTATIONS

We now discuss universal analog computation in the presence of symmetries. In analog quantum computation, universality means that the available Hamiltonians generate all traceless Hamiltonians on the Hilbert space, or equivalently that the corresponding dynamical Lie algebra is $\mathfrak{su}(\mathcal{H})$. When the available Hamiltonians preserve a symmetry, their DLA is contained in the invariant Lie algebra $\mathfrak{su}(\mathcal{H})^G$, and therefore cannot be universal by itself unless this invariant algebra already coincides with $\mathfrak{su}(\mathcal{H})$.

A natural way to restore universality is to add symmetry-breaking Hamiltonians. In this section, we give a sufficient condition under which such Hamiltonians, together with the invariant Lie algebra, generate the full algebra $\mathfrak{su}(\mathcal{H})$. This structure also motivates a framework for robust universal analog computation. The block encoding from Section VI B makes the invariant algebra act transversally on the encoded space, while the symmetry-breaking Hamiltonians are treated as resource Hamiltonians because they are not transversal for the block encoding.

A. Universal Analog Computations

a. Lie-algebraic universality. We first recall the standard notion of universal controllability in the analog setting. The propagator $U(t)$ satisfies

$$\dot{U}(t) = -iH(t)U(t), \qquad U(0) = I,$$

where $H(t)$ is defined in Eq. (53). The system is unitarily controllable if, for every target unitary $U_{\rm tar} \in SU(d)$, there exist a finite time T and control functions $u_k(t)$ such that

$$U(T) = U_{\rm tar}$$

up to a global phase.

The standard Lie-algebraic controllability criterion states that the system is universally controllable if and only if the dynamical Lie algebra generated by the available Hamiltonians is $\mathfrak{su}(d)$ [67]:

$$\mathrm{Lie}_{\mathbb{R}}\{iH_0, iH_1, \ldots, iH_m\} \cong \mathfrak{su}(d).$$

When this condition holds, we call the set $iH_0, iH_1, \ldots, iH_m$ universal. Controllability can also be formulated in terms of $U(d)$ rather than $SU(d)$, in which case the relevant Lie algebra is $\mathfrak{u}(d)$. Throughout this work, however, we consider controllability up to global phase. This is sufficient for closed-system quantum dynamics, since scalar Hamiltonian terms generate only global phases. We therefore work with traceless Hamiltonians and the Lie algebra $\mathfrak{su}(d)$.

b. Restoring universality by symmetry breaking. As discussed in Section VI A, if the drift Hamiltonian H_0 and the available global control Hamiltonians ${H_j}_{j=1}^m$ are invariant under a finite symmetry group G, then their DLA is contained in the invariant Lie algebra

$$\mathfrak{su}(\mathcal{H})^G \cong \mathfrak{su}(d_1) \oplus \cdots \oplus \mathfrak{su}(d_k) \oplus \mathfrak{u}(1)^{\oplus(r-1)}.$$

This algebra is generally a proper subalgebra of $\mathfrak{su}(\mathcal{H})$. Its precise block structure depends on the symmetry group G and on the representation of G on $\mathcal{H}$.

The question is whether universality can be restored by adding a small set of local symmetry-breaking Hamiltonians to the symmetric Hamiltonians. In principle, this is always possible by adding a full generating set of $\mathfrak{su}(\mathcal{H})$. Such a solution, however, does not use the structure of the symmetric system and is not useful as a design principle. The relevant question is instead how to choose a small and physically reasonable set of symmetry-breaking Hamiltonians that, together with the symmetric dynamics, generates the full algebra.

Related universality criteria for symmetry-breaking controls have been studied in Refs. [21, 22]. Here we use a sufficient condition tailored to the representation-theoretic decomposition induced by G. Let $\mathcal{G}^{\rm br}$ denote the set of added symmetry-breaking Hamiltonians, or equivalently the corresponding skew-Hermitian generators when inserted into the real Lie closure. To state the condition, we introduce two objects in Appendix S5: the coupling graph $\Gamma_{\rm co}$ and the Full First-Factor Span condition. The coupling graph records which symmetry sectors are connected by the breaking Hamiltonians, while the Full First-Factor Span condition ensures that these couplings are large enough to generate the required off-diagonal directions between sectors.

The resulting sufficient condition is the following.

Theorem 14. *If the coupling graph $\Gamma_{\rm co}$ is connected and the Full First-Factor Span condition holds for every edge in $\Gamma_{\rm co}$, then*

$$\mathrm{Lie}_{\mathbb{R}}\{\mathfrak{su}(\mathcal{H})^G \cup \mathcal{G}^{\rm br}\} = \mathfrak{su}(\mathcal{H}).$$

The proof and the precise definitions of $\Gamma_{\rm co}$ and the Full First-Factor Span condition are given in Appendix S5. In Appendix E.3, we also illustrate Theorem 14 on a three-qubit system with S_3 permutation symmetry. In that example, adding the single symmetry-breaking Hamiltonian

$$Z_1 + X_2$$

is sufficient to generate $\mathfrak{su}(\mathcal{H})$ together with the invariant algebra.

The theorem is stated using $\mathfrak{su}(\mathcal{H})^G$ rather than the actual DLA of the symmetric Hamiltonians. This distinction is important: the DLA generated by a particular set of symmetric Hamiltonians may be a proper subalgebra of $\mathfrak{su}(\mathcal{H})^G$, as discussed in Section VI A. The invariant algebra is nevertheless the natural object for the criterion, because it is determined directly by the representation theory of the G-action and is therefore easier to characterize. In the rest of this section, we focus on cases where the symmetric Hamiltonians generate the full invariant algebra $\mathfrak{su}(\mathcal{H})^G$. This assumption isolates the additional role of the symmetry-breaking Hamiltonians and is sufficient for the robust universal simulation framework considered below.

B. Breaking Hamiltonians as a resource

a. Transversal and non-transversal parts. Suppose that, for a given family of symmetric Hamiltonians $iH_0, iH_1, \ldots, iH_m$, we have chosen a set of symmetry-breaking Hamiltonians $\mathcal{G}^{\mathrm{br}}$ satisfying the universality condition of Theorem 14. We now describe how the covariant codes constructed above can be used as building blocks for this universal analog-control setting. The idea is to use a code space compatible with the continuous symmetry, i.e., the code space is preserved by the symmetry action, so the corresponding encoded operations can be implemented transversally whenever they arise from this symmetry. This provides a natural way to encode information while keeping as much of the relevant dynamics as possible in transversal form.

One option is to use a fully $\mathfrak{su}(d)$-covariant code, where $d = \dim \mathcal{H}$, described in previous sections. Such a code has the AQEC performance established in the previous sections and makes all generators of $\mathfrak{su}(d)$ act transversally. However, as discussed above, this approach can be costly, because the local physical dimension must scale with the dimension of the logical Hilbert space. This makes the fully covariant construction difficult to scale for large analog systems.

The block-encoding approach gives a more symmetry-adapted alternative. Instead of imposing covariance with respect to all of $\mathfrak{su}(\mathcal{H})$, we use a code covariant with respect to

$$\mathfrak{su}(d_1) \oplus \cdots \oplus \mathfrak{su}(d_k) \oplus \mathfrak{u}(1)^{\oplus(r-1)}.$$

For this code, the Hamiltonians in the invariant algebra $\mathfrak{su}(\mathcal{H})^G$ are implemented transversally. The symmetry-breaking Hamiltonians in $\mathcal{G}^{\mathrm{br}}$ are different: they do not preserve the symmetry-sector decomposition and therefore are not transversal for the block encoding. We therefore treat them as resource Hamiltonians. In general, these resource Hamiltonians may be nonlocal after encoding and may be harder to implement than the transversal invariant dynamics.

b. Three-qubit example. Consider again the example of three qubits with S_3 permutation symmetry. The corresponding invariant Lie algebra is

$$\mathfrak{su}(4) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1).$$

As shown in Example S4.6, the global control Hamiltonians H_x and H_y, together with the intrinsic Hamiltonian H_{zz} defined in Eq. (59), are block diagonal in the symmetry-adapted basis. By contrast, the symmetry-breaking Hamiltonian

$$Z_1 + X_2$$

does not preserve this block decomposition and is not block diagonal in the same basis; see Example S5.1. As shown in Ref. [16], the DLA generated by iH_x, iH_y, iH_{zz} coincides with $\mathfrak{su}(2^3)^{S_3}$. Therefore the enlarged set

$$\{iH_x, iH_y, iH_{zz}, i(Z_1 + X_2)\}$$

is universal.

This example also clarifies the role of the assumption that the symmetric DLA coincides with $\mathfrak{su}(\mathcal{H})^G$. For the purpose of universal logical computation, the available symmetric Hamiltonians do not have to arise from a fixed microscopic physical model. What is needed is a set of Hamiltonians that generates the desired invariant algebra. One may therefore choose a convenient finite symmetry group G and a corresponding generating set for $\mathfrak{su}(\mathcal{H})^G$, and then add symmetry-breaking Hamiltonians that restore universality. For example, in the case $G = S_N$, Ref. [16] gives explicit generators of $\mathfrak{su}(\mathcal{H})^{S_N}$:

$$iH_x = i\sum_{j=1}^{N} \sigma_x^{(j)}, \qquad iH_y = i\sum_{j=1}^{N} \sigma_y^{(j)},$$
$$iH_{zz} = i\sum_{1\le i<j\le N} \sigma_z^{(i)}\sigma_z^{(j)}.$$

In this example, the generators are also naturally motivated by physical collective-spin interactions.

c. Design trade-off. The design problem is therefore to choose the symmetry group G, equivalently the decomposition

$$\mathfrak{su}(\mathcal{H})^G \cong \mathfrak{su}(d_1) \oplus \cdots \oplus \mathfrak{su}(d_k) \oplus \mathfrak{u}(1)^{\oplus(r-1)},$$

together with a symmetry-breaking resource set $\mathcal{G}^{\mathrm{br}}$. These choices involve a trade-off. A larger symmetry group gives a smaller invariant algebra and therefore smaller blocks to encode covariantly. For instance, choosing $G = S_N$ makes $\mathfrak{su}(\mathcal{H})^G$ polynomial-dimensional in N, as in Eq. (60). However, imposing such a large symmetry can make the required symmetry-breaking resource set more complicated. Conversely, choosing a smaller symmetry group, such as $\mathbb{Z}_2$, gives less reduction in the invariant algebra, but universality may be restored with only one symmetry-breaking Hamiltonian; see Ref. [22].

Theorem 14 gives only a sufficient condition for universality. It does not determine the minimal number of symmetry-breaking Hamiltonians, nor does it characterize the simplest possible structure of such Hamiltonians. Understanding this optimization problem, including the trade-off between block-encoding efficiency and the complexity of the resource Hamiltonians, is left for future work.

VIII. DISCUSSION AND OUTLOOK

To summarize, in the first part of the paper, we constructed several families of $SU(d)$-covariant AQECC. The recovery error of these codes scales as $1/n$ for one-qudit, two-qudit, and three-qudit flagged erasure models, and for single-qudit erasure this scaling is near-optimal. We then constructed a decoder for the single-qudit flagged erasure model using the Petz recovery map. This decoder is near-optimal and achieves $1/n$ scaling, up to constants depending on the physical and logical dimensions. We also analyze the same codes under arbitrary flagged noise on at most three qudits, obtaining the more general, but weaker, scaling $1/\sqrt{n}$. Because exactly k-transitive permutation groups do not exist for $k > 3$, this symmetry-based construction cannot be extended directly beyond three erased sites. We removed this restriction by replacing exact permutation symmetry with a Haar-random multiplicity vector: a typical draw gives a code whose k-site reduced states are all close to one another, rather than exactly equal, for any fixed k, and we gave an efficient classical procedure for sampling such a typical code from a seed of $O(n \log d)$ random bits. This yields a family of $SU(d)$-covariant codes correctable against flagged erasure on any fixed number k of sites, with recovery error scaling as $O(\sqrt{k(d^2-1)}/n)$. The trace-norm argument used for non-erasure noise applies to these codes as well, giving $O(\sqrt{k/n})$ correctability against arbitrary flagged noise on any fixed number k of sites. We read this less as the construction of one more code than as a statement about the covariant sector itself. Once covariance has fixed the isotypic component, the only remaining freedom is a direction in the multiplicity space, and the directions along which some erased subsystem learns an atypically large amount about the logical state have exponentially small Haar measure. Badly aligned directions exist, the unencoded seed vector among them, but they are non-generic, so approximate decoupling from any fixed-size erased subsystem is typical rather than fine-tuned. The efficient sampler adds a second reading of the same kind. Haar randomness establishes that good directions are abundant, but a generic one is not something that can be written down, and the sampling procedure shows that the abundance survives when the continuous ensemble is replaced by four-wise independent phases from a seed of $O(n \log d)$ bits. What the error-correcting property depends on is therefore not Haar randomness but the low-order moments the concentration argument consumes, and those are already available inside a structured describable family.

In the second part of the paper, we introduced block encodings that are covariant with respect to algebras of the form $\mathfrak{su}(d_1) \oplus \cdots \oplus \mathfrak{su}(d_k) \oplus \mathfrak{u}(1)^{\oplus(r-1)}$. These encodings are motivated by quantum simulation, where the relevant dynamics are determined by the DLA of the available Hamiltonians. We analyzed the resulting block codes under two-qudit erasure noise, including both in-block and inter-block erasures, and obtained $1/n$ scaling of the recovery error.

In the third part of the paper, we proposed a framework for robust universal analog computation. We first gave a sufficient condition under which adding a set of symmetry-breaking Hamiltonians to the invariant Lie subalgebra generates a universal DLA. When combined with the block encoding developed earlier, the invariant subalgebra acts transversally on the encoded space, while the symmetry-breaking Hamiltonians become non-transversal resource Hamiltonians needed to complete universality. This suggests a possible route toward fault-tolerant analog computation in which a large symmetry-preserving sector is protected by covariance, while a controlled set of non-transversal resource Hamiltonians supplies universality.

There is an interesting resemblance between the $SU(d)$-covariant AQECC codes constructed here and stabilizer codes. First, for stabilizer codes, correcting flagged k-qubit erasures is equivalent to correcting arbitrary k-qubit errors at known locations. A similar mechanism appears in our setting. Indeed, the complementary channel for arbitrary flagged local noise factors through the reduced encoded state: it is obtained by first taking the local reduced state, which is precisely the information seen by the environment in the erasure model, and then applying the complementary channel of the local noise. Since this second step is linear, the coefficient suppressing the logical-state-dependent part of the reduced state is preserved. Thus, the same reduced-state structure that makes the code good against erasures also implies approximate correctability against arbitrary flagged local noise, although the general trace-norm argument gives a weaker $1/\sqrt{n}$ bound unless the fidelity can be evaluated more sharply.

Second, there is an analogy with the behavior of tensor products of stabilizer codes. For stabilizer codes, taking a tensor product of two codes with the same distance preserves the distance scale. In our setting, the block encoding behaves similarly in terms of asymptotic behavior. Namely, the $\mathfrak{su}(d_1) \oplus \mathfrak{su}(d_2)$-covariant block code is obtained from the corresponding $SU(d_1)$- and $SU(d_2)$-covariant approximate codes, and its performance under two-qudit erasure noise preserves the same $1/n$ scaling. This remains true for both in-block and inter-block erasures.

Although continuous covariance imposes an $\Omega(1/n)$ lower bound on the worst-case recovery distance, this scaling should be viewed as a resource–accuracy trade-off rather than as an accuracy floor. A target error ε can still be reached with $n = O(\varepsilon^{-1})$ physical subsystems. Such inverse-polynomial overhead can be compatible with quantum-simulation applications, particularly when the objective is to reproduce local or few-body observables over finite evolution times rather than the entire many-body state with extremely high global fidelity. Indeed, broad classes of finite-time local-observable simulation tasks are known to be stable against local implementation errors [32]. The inverse-linear scaling also has a natural connection to quantum metrology, as metrological bounds on quantum Fisher information provide one route to approximate Eastin–Knill inequalities [43, 54], and the $d = \Theta(1/n)$ scaling in our fidelity-based distance corresponds to an infidelity $1-F = \Theta(1/n^2)$, reminiscent of Heisenberg scaling. This connection does not identify recovery error with metrological estimation error, but it emphasizes that the symmetry constraint imposes a fundamental scaling law whose cost can be systematically reduced by increasing the available physical resources.

IX. FUTURE WORK

Several important directions remain open. As discussed above, it is not necessary for all k-qudit reduced states to be identical, even up to the identification of tuples of sites, in order to distribute logical information sufficiently uniformly across the physical system: more complicated distributions of reduced states can still lead to good AQEC performance. In Section V C we used this observation to construct a covariant AQECC with good scaling under general k-qudit erasure noise, using the reduced-state method developed in this work: replacing the exact permutation symmetry, which does not exist for $k > 3$, by a Haar-random, efficiently sampled multiplicity vector gives a code whose k-site reduced states are all close to a common mean, with recovery error scaling as $\frac{\sqrt{k(d^2-1)}}{n}$ for fixed k, confirming the $\sqrt{k}/n$ scaling expected from the reduced-state method. Two related questions remain open: whether the same scaling can be reached by a code with no probability of failure, and whether the constant obtained here is optimal.

Another natural direction is to analyze recovery maps more explicitly. The Petz, or transpose, recovery map can be generalized directly to the k-qudit erasure setting. However, understanding its exact performance remains an important open problem. In this work, we establish near-optimality at the level of scaling in n, but we do not determine the sharp constants depending on the physical representation, the logical dimension. A more detailed analysis of the Petz recovery map may therefore clarify whether it is optimal beyond the level of asymptotic scaling.

Our analysis of arbitrary local noise also generalizes formally to k-qudit noise, provided the corresponding k-qudit reduced states have the appropriate form. In particular, if one constructs an encoding for which the k-qudit reduced states approach a fixed input-independent state as $n \to \infty$, then an $O_{d,k}(n^{-1/2})$ correctability bound for arbitrary flagged k-qudit noise follows essentially automatically from the trace-norm argument. However, this bound is likely not sharp. In contrast to erasure channels, arbitrary local noise channels are not generally covariant, which makes a direct fidelity calculation substantially harder, see [43, 48]. Improving the $O_{d,k}(n^{-1/2})$ estimate, and understanding whether $O_{d,k}(n^{-1})$ scaling can be recovered for broader classes of local noise, remains an important question.

It would also be interesting to consider more general choices of the individual physical representation. In this work, we mostly focus on irreducible physical subsystems, such as V_{ω_r}. However, there are natural covariant codes whose individual physical systems are reducible. For example, the W-state-type code [42, 48] uses a physical subsystem carrying a direct sum of the fundamental representation and a trivial representation, $V_{\omega_1} \oplus V_0$. It would be useful to understand the broader family of codes to which this example belongs and to analyze such codes under general noise models.

There are also several open questions motivated by quantum simulation. For a fixed physical system, one would like to find an encoding adapted to its actual DLA. In general, the DLA need not have the block form $\mathfrak{su}(d_1) \oplus \cdots \oplus \mathfrak{su}(d_k) \oplus \mathfrak{u}(1)^{\oplus(r-1)}$. Thus, if the goal is to protect a given analog simulation, it may be necessary to construct covariant AQECC for other Lie algebras, such as $\mathfrak{so}(d)$, $\mathfrak{sp}(d)$, or other Lie algebras that can arise as DLA of physical systems, see [69, 70].

Finally, many questions remain open on the side of universal analog computation. Even without encoding, it is important to understand the necessary conditions under which a set of symmetry-breaking Hamiltonians, together with the invariant Lie subalgebra, generates the full algebra required for universality. Ideally, one would like a complete criterion giving necessary and sufficient conditions. From the encoding perspective, the goal is to find both an efficient encoding and a suitable set of breaking Hamiltonians: the encoding should minimize the overhead, while the breaking Hamiltonians should be as physically reasonable as possible, for example not too nonlocal and not too numerous. Understanding this tradeoff is a necessary step toward the longer-term goal of establishing a threshold theorem for such codes and developing a framework for fault-tolerant analog quantum computation.

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DATA AVAILABILITY

No data were created or analyzed in this study.

[1] F. Arute, K. Arya, R. Babbush, D. Bacon, J. C. Bardin, R. Barends, R. Biswas, S. Boixo, F. G. S. L. Brandao, D. A. Buell, *et al.*, Quantum supremacy using a programmable superconducting processor, Nature **574**, 505 (2019).

[2] Y. Kim, A. Eddins, S. Anand, K. X. Wei, E. van den Berg, S. Rosenblatt, H. Nayfeh, Y. Wu, M. Zaletel, K. Temme, and A. Kandala, Evidence for the utility of quantum computing before fault tolerance, Nature **618**, 500 (2023).

[3] A. G. Fowler and C. Gidney, Low overhead quantum computation using lattice surgery (2018), arXiv:1808.06709 [quant-ph].

[4] C. Gidney and M. Ekerå, How to factor 2048 bit RSA integers in 8 hours using 20 million noisy qubits, Quantum **5**, 433 (2021), arXiv:1905.09749.

[5] I. Bloch, J. Dalibard, and S. Nascimbène, Quantum simulations with ultracold quantum gases, Nat. Phys. **8**, 267 (2012).

[6] A. Browaeys and T. Lahaye, Many-body physics with individually controlled Rydberg atoms, Nature Physics **16**, 132 (2020).

[7] A. J. Daley, I. Bloch, C. Kokail, S. Flannigan, N. Pearson, M. Troyer, and P. Zoller, Practical quantum advantage in quantum simulation, Nature **607**, 667 (2022).

[8] F. Schäfer, T. Fukuhara, S. Sugawa, Y. Takasu, and Y. Takahashi, Tools for quantum simulation with ultracold atoms in optical lattices, Nat. Rev. Phys. **2**, 411 (2020).

[9] M. Xu, L. H. Kendrick, A. Kale, Y. Gang, C. Feng, S. Zhang, A. W. Young, M. Lebrat, and M. Greiner, A neutral-atom Hubbard quantum simulator in the cryogenic regime, Nature **642**, 909 (2025).

[10] A. Mazurenko, C. S. Chiu, G. Ji, M. F. Parsons, M. Kanász-Nagy, R. Schmidt, F. Grusdt, E. Demler, D. Greif, and M. Greiner, A cold-atom Fermi–Hubbard antiferromagnet, Nature **545**, 462 (2017).

[11] H.-J. Shao, Y.-X. Wang, D.-Z. Zhu, Y.-S. Zhu, H.-N. Sun, S.-Y. Chen, C. Zhang, Z.-J. Fan, Y. Deng, X.-C. Yao, Y.-A. Chen, and J.-W. Pan, Antiferromagnetic phase transition in a 3D fermionic Hubbard model, Nature **632**, 267 (2024).

[12] T. Chalopin, P. Bojović, S. Wang, T. Franz, A. Sinha, Z. Wang, D. Bourgund, J. Obermeyer, F. Grusdt, A. Bohrdt, L. Pollet, A. Wietek, A. Georges, T. Hilker, and I. Bloch, Observation of emergent scaling of spin–charge correlations at the onset of the pseudogap, Proceedings of the National Academy of Sciences **123**, 10.1073/pnas.2525539123 (2026).

[13] M. Tavis and F. W. Cummings, Exact solution for an n-molecule—radiation-field hamiltonian, Phys. Rev. **170**, 379 (1968).

[14] D. D'Alessandro, *Introduction to Quantum Control and Dynamics*, 2nd ed. (Chapman and Hall/CRC, 2021).

[15] J. Chen, Y. Zhou, J. Bian, J. Li, and X. Peng, Subspace controllability of symmetric spin networks, Phys. Rev. A **102**, 032602 (2020).

[16] F. Albertini and D. D'Alessandro, Controllability of symmetric spin networks, Journal of Mathematical Physics **59**, 052102 (2018).

[17] F. Albertini and D. D'Alessandro, Subspace controllability of bipartite symmetric spin networks, Linear Algebra and its Applications **585**, 1 (2020).

[18] X. Wang, D. Burgarth, and S. Schirmer, Subspace controllability of spin-$\frac{1}{2}$ chains with symmetries, Phys. Rev. A **94**, 052319 (2016).

[19] X. Wang, P. Pemberton-Ross, and S. G. Schirmer, Symmetry and subspace controllability for spin networks with a single-node control, IEEE Transactions on Automatic Control **57**, 1945 (2012).

[20] P. Zanardi and S. Lloyd, Universal control of quantum subspaces and subsystems, Phys. Rev. A **69**, 022313 (2004).

[21] F. Albertini and D. D'Alessandro, Quantum subspace controllability implying full controllability, Linear Algebra and its Applications **705**, 207 (2025).

[22] H.-Y. Hu, A. M. Gomez, L. Chen, A. Trowbridge, A. J. Goldschmidt, Z. Manchester, F. T. Chong, A. Jaffe, and S. F. Yelin, Universal dynamics with globally controlled analog quantum simulators (2025), arXiv:2508.19075 [quant-ph].

[23] S. Krinner, N. Lacroix, A. Remm, A. Di Paolo, E. Genois, C. Leroux, C. Hellings, S. Lazar, F. Swiadek, J. Herrmann, *et al.*, Realizing repeated quantum error correction in a distance-three surface code, Nature **605**, 669 (2022).

[24] Google Quantum AI and Collaborators, Quantum error correction below the surface code threshold, Nature **638**, 920 (2025).

[25] C. Ryan-Anderson, J. G. Bohnet, K. Lee, D. Gresh, A. Hankin, J. P. Gaebler, D. Francois, A. Chernoguzov, D. Lucchetti, N. C. Brown, T. M. Gatterman, S. K. Halit, K. Gilmore, J. A. Gerber, B. Neyenhuis, D. Hayes, and R. P. Stutz, Realization of real-time fault-tolerant quantum error correction, Phys. Rev. X **11**, 041058 (2021).

[26] A. Paetznick, M. P. da Silva, C. Ryan-Anderson, J. M. Bello-Rivas, J. P. C. III, A. Chernoguzov, J. M. Dreiling, C. Foltz, F. Frachon, J. P. Gaebler, T. M. Gatterman, L. Grans-Samuelsson, D. Gresh, D. Hayes, N. Hewitt, C. Holliman, C. V. Horst, J. Johansen, D. Lucchetti, Y. Matsuoka, M. Mills, S. A. Moses, B. Neyenhuis, A. Paz, J. Pino, P. Siegfried, A. Sundaram, D. Tom, S. J. Wernli, M. Zanner, R. P. Stutz, and K. M. Svore, Demonstration of logical qubits and repeated error correction with better-than-physical error rates (2024), arXiv:2404.02280 [quant-ph].

[27] B. W. Reichardt, D. Aasen, R. Chao, A. Chernoguzov, W. van Dam, J. P. Gaebler, D. Gresh, D. Lucchetti, M. Mills, S. A. Moses, B. Neyenhuis, A. Paetznick, A. Paz, P. E. Siegfried, M. P. da Silva, K. M. Svore, Z. Wang, and M. Zanner, Demonstration of quantum computation and error correction with a tesseract code (2024), arXiv:2409.04628 [quant-ph].

[28] D. Bluvstein, S. J. Evered, A. A. Geim, S. H. Li, H. Zhou, T. Manovitz, S. Ebadi, M. Cain, M. Kalinowski, D. Hangleiter, J. P. B. Ataides, N. Maskara, I. Cong, X. Gao, P. S. Rodriguez, T. Karolyshyn, G. Semeghini, M. J. Gullans, M. Greiner, V. Vuletić, and M. D. Lukin, Logical quantum processor based on reconfigurable atom arrays, Nature **626**, 58 (2024).

[29] B. W. Reichardt, A. Paetznick, D. Aasen, I. Basov, J. M. Bello-Rivas, P. Bonderson, R. Chao, W. van Dam, M. B. Hastings, R. V. Mishmash, A. Paz, M. P. da Silva, A. Sundaram, K. M. Svore, A. Vaschillo, Z. Wang, M. Zanner, W. B. Cairncross, C.-A. Chen, D. Crow, H. Kim, J. M. Kindem, J. King, M. McDonald, M. A. Norcia, A. Ryou, M. Stone, L. Wadleigh, K. Barnes, P. Battaglino, T. C. Bohdanowicz, G. Booth, A. Brown, M. O. Brown, K. Cassella, R. Coxe, J. M. Epstein, M. Feldkamp, C. Griger, E. Halperin, A. Heinz, F. Hummel, M. Jaffe, A. M. W. Jones, E. Kapit, K. Kotru, J. Lauigan, M. Li, J. Marjanovic, E. Megidish, M. Meredith, R. Morshead, J. A. Muniz, S. Narayanaswami, C. Nishiguchi, T. Paule, K. A. Pawlak, K. L. Pudenz, D. R. Pérez, J. Simon, A. Smull, D. Stack, M. Urbanek, R. J. M. van de Veerdonk, Z. Vendeiro, R. T. Weverka, T. Wilkason, T.-Y. Wu, X. Xie, E. Zalys-Geller, X. Zhang, and B. J. Bloom, Fault-tolerant quantum computation with a neutral atom processor (2025), arXiv:2411.11822 [quant-ph].

[30] M. Reiher, N. Wiebe, K. M. Svore, D. Wecker, and M. Troyer, Elucidating reaction mechanisms on quantum computers, Proceedings of the National Academy of Sciences **114**, 7555 (2017).

[31] I. D. Kivlichan, C. Gidney, D. W. Berry, N. Wiebe, J. McClean, W. Sun, Z. Jiang, N. Rubin, A. Fowler, A. Aspuru-Guzik, H. Neven, and R. Babbush, Improved fault-tolerant quantum simulation of condensed-phase correlated electrons via trotterization, Quantum **4**, 296 (2020), arXiv:1902.10673.

[32] R. Trivedi, A. F. Rubio, and J. I. Cirac, Quantum advantage and stability to errors in analogue quantum simulators, Nature Communications **15**, 6507 (2024).

[33] V. Kashyap, G. Styliaris, S. Mouradian, J. I. Cirac, and R. Trivedi, Accuracy guarantees and quantum advantage in analog open quantum simulation with and without noise, Phys. Rev. X **15**, 021017 (2025).

[34] R.-C. Guo, Y. Gu, and D. E. Liu, Mitigating errors in analog quantum simulation by hamiltonian reshaping or hamiltonian rescaling, npj Quantum Information **11**, 14 (2025).

[35] Y. Cao, S. Liu, H. Deng, Z. Xia, X. Wu, and Y.-X. Wang, Robust analog quantum simulators by quantum error-detecting codes (2024), arXiv:2412.07764.

[36] J. Atalaya, S. Zhang, M. Y. Niu, A. Babakhani, H. C. H. Chan, J. M. Epstein, and K. B. Whaley, Continuous quantum error correction for evolution under time-dependent hamiltonians, Phys. Rev. A **103**, 042406 (2021).

[37] B. Eastin and E. Knill, Restrictions on transversal encoded quantum gate sets, Physical Review Letters **102**, 110502 (2009).

[38] P. Hayden, S. Nezami, S. Popescu, and G. Salton, Error correction of quantum reference frame information, PRX Quantum **2**, 010326 (2021).

[39] P. Gupta, A. Morello, and B. C. Sanders, Universal transversal gates (2024), arXiv:2410.07045 [quant-ph].

[40] B. Schumacher and M. D. Westmoreland, Approximate quantum error correction, Quantum Information Processing **1**, 5 (2002), arXiv:quant-ph/0112106 [quant-ph].

[41] C. Bény and O. Oreshkov, General conditions for approximate quantum error correction and near-optimal recovery channels, Phys. Rev. Lett. **104**, 120501 (2010).

[42] P. Faist, S. Nezami, V. V. Albert, G. Salton, F. Pastawski, P. Hayden, and J. Preskill, Continuous symmetries and approximate quantum error correction, Phys. Rev. X **10**, 041018 (2020).

[43] S. Zhou, Z.-W. Liu, and L. Jiang, New perspectives on covariant quantum error correction, Quantum **5**, 521 (2021), arXiv:2005.11918.

[44] P. Faist, M. P. Woods, V. V. Albert, J. M. Renes, J. Eisert, and J. Preskill, Time-energy uncertainty relation for noisy quantum metrology, PRX Quantum **4**, 040336 (2023).

[45] L. Kong and Z.-W. Liu, Near-optimal covariant quantum error-correcting codes from random unitaries with symmetries, PRX Quantum **3**, 020314 (2022).

[46] Y. Yang, Y. Mo, J. M. Renes, G. Chiribella, and M. P. Woods, Optimal universal quantum error correction via bounded reference frames, Phys. Rev. Res. **4**, 023107 (2022).

[47] C.-J. Lin, Z.-W. Liu, V. V. Albert, and A. V. Gorshkov, Covariant quantum error-correcting codes with metrological entanglement advantage, Phys. Rev. Lett. **135**, 110801 (2025).

[48] R. Alexander, A new approximate eastin-knill theorem, arXiv preprint arXiv:2505.00427 10.48550/arXiv.2505.00427 (2025), arXiv:2505.00427.

[49] J. Yi, W. Ye, D. Gottesman, and Z.-W. Liu, Complexity and order in approximate quantum error-correcting codes, Nature Physics **20**, 1798 (2024).

[50] A. Almheiri, X. Dong, and D. Harlow, Bulk locality and quantum error correction in AdS/CFT, Journal of High Energy Physics **2015**, 163 (2015).

[51] F. Pastawski, B. Yoshida, D. Harlow, and J. Preskill, Holographic quantum error-correcting codes: Toy models for the bulk/boundary correspondence, Journal of High Energy Physics **2015**, 149 (2015).

[52] D. Harlow and H. Ooguri, Constraints on symmetries from holography, Phys. Rev. Lett. **122**, 191601 (2019).

[53] M. P. Woods and Á. M. Alhambra, Continuous groups of transversal gates for quantum error correcting codes from finite clock reference frames, Quantum **4**, 245 (2020).

[54] A. Kubica and R. Demkowicz-Dobrzański, Using quantum metrological bounds in quantum error correction: A simple proof of the approximate eastin-knill theorem, Phys. Rev. Lett. **126**, 150503 (2021).

[55] D. PETZ, Sufficiency of channels over von neumann algebras, The Quarterly Journal of Mathematics **39**, 97 (1988), https://academic.oup.com/qjmath/article-pdf/39/1/97/4559225/39-1-97.pdf.

[56] H. Barnum and E. Knill, Reversing quantum dynamics with near-optimal quantum and classical fidelity, Journal of Mathematical Physics **43**, 2097 (2002).

[57] H. K. Ng and P. Mandayam, Simple approach to approximate quantum error correction based on the transpose channel, Phys. Rev. A **81**, 062342 (2010).

[58] See Supplemental Material at [URL will be inserted by publisher] for the nomenclature, detailed proofs, reduced-state calculations, the explicit recovery-map construction, and the analysis of erasure on an arbitrary fixed number of qudits.

[59] A. S. Fletcher, P. W. Shor, and M. Z. Win, Optimum quantum error recovery using semidefinite programming, Phys. Rev. A **75**, 012338 (2007).

[60] A. Denys and A. Leverrier, Quantum error-correcting codes with a covariant encoding, Phys. Rev. Lett. **133**, 240603 (2024).

[61] B. C. Hall, *Lie Groups, Lie Algebras, and Representations: An Elementary Introduction*, 2nd ed., Graduate Texts in Mathematics, Vol. 222 (Springer Cham, 2015).

[62] D. Bacon, I. L. Chuang, and A. W. Harrow, Efficient quantum circuits for schur and clebsch-gordan transforms, Phys. Rev. Lett. **97**, 170502 (2006).

[63] R. Goodman and N. R. Wallach, *Symmetry, Representations, and Invariants*, 1st ed., Graduate Texts in Mathematics, Vol. 255 (Springer New York, 2009).

[64] W. Fulton and J. Harris, *Representation Theory: A First Course*, Graduate Texts in Mathematics, Vol. 129 (Springer New York, 1991).

[65] D. Gorenstein, R. Lyons, and R. Solomon, *The Classification of the Finite Simple Groups*, Mathematical Surveys and Monographs, Vol. 40 (American Mathematical Society, 1994).

[66] J. D. Dixon and B. Mortimer, *Permutation Groups*, Graduate Texts in Mathematics, Vol. 163 (Springer, 1996).

[67] F. Albertini and D. D'Alessandro, Notions of controllability for bilinear multilevel quantum systems, IEEE Transactions on Automatic Control **48**, 1399 (2003).

[68] D. D'Alessandro and J. T. Hartwig, Dynamical decomposition of bilinear control systems subject to symmetries, J Dyn Control Syst **27**, 1 (2021), arXiv:1806.01179v2.

[69] R. Wiersema, E. Kökcü, A. F. Kemper, and B. N. Bakalov, Classification of dynamical lie algebras of 2-local spin systems on linear, circular and fully connected topologies, npj Quantum Information **10**, 110 (2024).

[70] E. Kökcü, R. Wiersema, A. F. Kemper, and B. N. Bakalov, Classification of dynamical lie algebras generated by spin interactions on undirected graphs, arXiv preprint arXiv:2409.19797 10.48550/arXiv.2409.19797 (2024), arXiv:2409.19797.